The Climate Credit Channel: Physical Shocks and Monetary Policy Transmission in South Africa

Tichinashe Mabugu* Xolani Sibande†

October 27, 2026

AAAAAA

Keywords:
JEL Codes:

*Corresponding Author. Economic Research Department, South African Reserve Bank; Email: xolani.sibande@resbank.co.za
†Economic Research Department, South African Reserve Bank; Email: tichinashe.mabugu@resbank.co.za.

1 Introduction

The credit channel is a central mechanism through which monetary policy reaches the real economy. By altering the external finance premium — the wedge between the cost of internal and external funds generated by credit-market frictions — monetary policy shifts the borrowing costs and balance sheet positions of both firms and households, with downstream effects on investment, consumption, and ultimately output and prices (Bernanke and Gertler, 1995). Two linkages underpin this transmission: the balance sheet channel, which operates through changes in borrower net worth and collateral values, and the bank lending channel, which operates through the supply of loanable funds (Hall, 2001). Considerable empirical evidence has established the balance sheet channel as quantitatively important in advanced economies (Aysun and Hepp, 2013; Altavilla et al., 2024; Ashcraft and Campello, 2007), yet a full accounting of its strength remains elusive, particularly in emerging markets where the institutional context differs markedly from the settings in which most of this evidence was gathered (Pirozhkova and Viegi, 2025).

A related challenge is that observed credit responses to monetary policy do not always conform to the predictions of standard transmission models. Information effects, heterogeneous bank balance sheets, and the distinction between anticipated and unanticipated rate changes all generate dispersion in the pass-through of policy to credit conditions (Altavilla et al., 2020; Kleimeier and Sander, 2006). This motivated the development of structural monetary policy surprise measures — residuals purged of the component of rate changes that can be predicted from the central bank's own information set, or from financial market expectations — which isolate the genuinely unexpected component of policy (Romer and Romer, 2004; Miyajima, 2024). It is precisely this unexpected component, by definition unconfounded by the macroeconomic conditions that prompted the rate decision, that permits causal inference about the credit channel. Yet even with carefully identified surprise measures, the literature documents atypical and heterogeneous responses that standard models struggle to rationalise, suggesting that the full set of factors shaping credit channel strength is not yet well understood.

This paper proposes that physical climate shocks constitute one such factor. Climate shocks

— deviations of temperature and precipitation from their long-run seasonal norms — affect firms and households through channels that closely mirror the determinants of the external finance premium: they disrupt production, damage capital and collateral, compress profitability, and erode net worth (Dafermos et al., 2018; Aglietta and Espagne, 2016). If such damage coincides with a monetary policy tightening, the deterioration in borrower balance sheets may amplify the credit response — lending rates rising further or credit volumes contracting more sharply than the policy shock alone would produce. Alternatively, if a benign climate shock supports economic activity at the same moment that policy tightens, it may cushion the credit response and dampen the effectiveness of monetary policy. The direction and magnitude of this interaction is an empirical question, and one that has received limited attention despite its clear implications for how central banks should interpret the transmission of their decisions.

The specific question this paper investigates is: does the coincidence of physical climate shocks with monetary policy surprises alter the strength and direction of credit channel transmission in South Africa? Concretely, when a temperature or precipitation shock accompanies an unexpected tightening, are lending rates kept higher for longer, or is the rate response muted? Do credit volumes contract more sharply, or does climate-driven demand for borrowing offset the policy impulse? We examine both amplification — in which climate shocks strengthen the credit response to monetary policy — and dampening — in which they weaken it — across three monetary policy surprise measures, two climate shock variables, and twelve disaggregated credit series covering corporate and household borrowers at the bank level. South Africa provides a particularly informative setting: it is a climate-vulnerable emerging economy with well-developed financial markets, a credible independent central bank, and granular bank-level credit data disaggregated by borrower type and product that is rarely available in comparable economies.

Using bank-level panel data over 2011–2025 and a combination of panel fixed effects regressions and panel local projections, we find that the interaction between climate and monetary policy shocks operates primarily through the price of credit — lending rates — rather than its quantity, consistent with a risk-repricing mechanism. The effects are heterogeneous across shock types and credit products, with amplification and dampening both present depending

on the specific climate and monetary policy pairing. They are also persistent, with the interaction peaking at 8–10 quarters after the shock and most pronounced under extreme climate conditions at the tails of the distribution. These findings point to physical climate risk as a systematic but previously underappreciated source of variation in the strength of the credit channel, with implications for how climate-vulnerable central banks model and communicate monetary policy transmission.

The remainder of the paper is organised as follows. Section 2 develops the conceptual framework linking climate shocks to the external finance premium. Section 3 describes the data and methodology, including the construction of the monetary policy surprise and climate shock series. Section 4 presents the main fixed effects and local projection results. Section 5 examines heterogeneity across shock types, credit products, and climate intensity. Section 6 concludes.

2 Conceptual Framework

We argue that climate shocks may alter the strength of the credit channel by affecting the financial position of corporates, households and banks. For simplicity, we illustrate the transmission of a contractionary monetary policy shock through the credit channel throughout this section. We distinguish between two possible effects that can ensue when a monetary policy shock coincides with a climate shock: amplification and dampening. We define amplification as a strengthening of the response of corporates or households to a monetary policy shock, such that the response of credit rates or volumes moves in the same direction as the response to the monetary policy shock alone, but with a larger absolute magnitude when the climate shock is present. A dampening effect, in contrast, occurs when the response continues to move in the same direction as the response to the monetary policy shock alone, but with a smaller absolute magnitude when the climate shock is present.

2.1 Corporates

From the perspective of corporates, we draw from a framework put forward by Bean et al. (2002) as follows:

$$R_{it}^{c} = R_t + f(D_t/E_t), \qquad f' > 0 \tag{1}$$

where R_{it}^{c} is the interest rate faced by corporates which is dependent on R_t, the risk-free interest rate and a function of the corporate's financial position, in this case represented by its Debt (D_t) to Equity (E_t) ratio. This function, f therefore represents the external finance premium. Insofar as there are information asymmetries or financial frictions, Equation 1 holds.

We posit that climate shocks alter the interest rate faced by corporates through changes in their financial position, which affect the external finance premium and, consequently, the cost of borrowing (Laumer and Schaffer, 2025). In particular, a climate shock can yield supply chain disruptions, damage of capital and production losses. These effects may reduce revenue and profits. Simultaneously, climate shock-related damage may reduce the value of collateral against which corporates borrow. These effects can increase corporate debt, D_t, while the associated decline in profitability, asset values and net worth can reduce equity, E_t. Consequently, a climate shock may increase a corporate's finance premium and in turn, raise the cost of external finance faced by the corporate. In this way, an adverse climate shock amplifies a contractionary monetary policy shock.

The same framework allows for a dampening effect of contractionary monetary policy. That is, firms with large cash buffers, liquid assets or adequate insurance coverage may experience little or no deterioration in their balance sheets, resulting in little or no change in D_t/E_t and consequently in the external finance premium, $f(D_t/E_t)$, following a climate shock. This resilience can also render corporates less responsive to the higher borrowing costs. Thus, climate shocks need not strengthen the transmission of monetary policy and may instead leave it unchanged or weaken it. Evidence that cash holdings can mitigate the effect of monetary tightening on corporate investment supports this mechanism (Sharpe and Suarez, 2020).

A final, related potential propagation mechanism for corporates arises from the possibility that corporate borrowing and investment decisions may be relatively less responsive to changes

in borrowing costs. If climate shocks demand a need for adaptation, replacement of capital or reconstruction, corporates may have a strong incentive to undertake this investment even when R_t rises. In this case, corporates may become less sensitive to the cost of external finance. This does not necessarily mean that R_{it}^c itself falls; rather, the higher financing cost has a smaller effect on corporates' borrowing decisions. This effect would be most evident when higher interest rates coincide with an increase in credit volumes. Consistent with this possibility, Xu et al. (2025) demonstrate that adaptation policy increases firm investment even in the absence of an easing of financial constraints faced by firms.

2.2 Households

Analogous to Equation 1, we this of households facing an interest rate according to the following Equation:

$$R_{it}^h = R_t + \phi_t, \qquad \phi_t > 0 \tag{2}$$

where R_{it}^h is the interest rate faced by households as a result of the policy rate R_t and the household-specific spread between the policy rate and the cost of borrowing ϕ_t. The spread can be viewed as reflecting household creditworthiness, indebtedness, income and collateral and other characteristics that affect lenders' assessment of credit risk and the terms at which credit is supplied. Thus, ϕ_t allows households facing the same monetary policy rate to nevertheless face different borrowing costs. This allowance for differentiated credit market outcomes faced by households is driven by evidence suggesting that transmission of monetary policy varies across households according to a range of factors including their exposure to income fluctuations (Slacalek et al., 2020).

The household mechanism ensues similarly to the case of firms. On the one hand, climate shocks may amplify the credit channel's response to monetary policy shocks. It follows that a climate shock may reduce household income, wealth or damage collateral. This deterioration in the balance sheet of households increases the household-specific spread, ϕ_t, causing affected households to face higher borrowing costs for a given monetary policy rate. On the other hand, climate shocks may dampen this impact in a similar way to the case of corporates. That

is households with liquid assets, insurance coverage or other financial buffers may be able to absorb climate-related losses without a deterioration in their financial position or increased reliance on external finance. In such cases, the climate shock may generate little change in ϕ_t and therefore need not amplify the response of household borrowing costs to monetary policy.Empirical evidence suggests that insurance may lower debt levels in the face of disasters as households use insurance payouts to pay off mortgages rather than to reconstruct (Gallagher and Hartley, 2017).

2.3 Banks

The household and firm mechanisms are connected through the financial sector. Climate shocks can affect not only borrowers' demand for credit but also banks' willingness to supply it. There is a possibility that during climate shock episodes, creditors heighten the importance they attach to corporates' and households' balance sheet strength. Indeed, there is evidence that mortgage arreas is higher in regions that have faced more damage from climate shocks relative to less damaged regions (Ho et al., 2023) Recent evidence suggests that in response, banks pose stricter lending conditions to borowers in disaster-prone regions (Duanmu et al., 2022).

On the other hand, a climate shock may dampen the transmission of monetary policy if it reduces the demand for credit. A deterioration in economic activity following a climate shock or adequate buffers to climate shocks may weaken corporates' and households' demand for new borrowing, reducing banks' loan demand and limiting the extent to which the shock translates into tighter credit conditions. When loan demand is weaker, banks may have less scope or incentive to adjust lending rates in response to changes in the riskiness of their borrowers. In this case, the climate shock may generate little additional increase in the spread between lending rates and the policy rate, thereby limiting its potential to amplify the increase in borrowing costs associated with a monetary policy tightening.

2.4 Policy

We must also consider the role that the policy rate, itself, plays in the determination of credit rates and, by extension, volumes. Two related concepts influence our framework. The first is around the mechanics of monetary policy transmission. Monetary policy transmission to lending rates is notoriously heterogeneous across banks and dependent on several factors such as bank balance sheet characteristics (Altavilla et al., 2020).

The second concept relates to the distinction between an anticipated interest rate change and an unanticipated rate change. Monetary policy passthrough in the credit market is faster when the policy rate is correctly anticipated (Kleimeier and Sander, 2006). We study monetary policy surprises which, by design, are unanticipated.

This has non-negligible implications for credit condition outcomes. In particular,…

2.5 Climate Shock Non-Linearity

Until now we have assumed that climate shocks yield adverse effects on corporates, households and banks. This is not necessarily always the case. Consider the case of precipitation shocks. One of the main ways that climate shocks impact macroeconomic outcomes is through the agricultural sector. Agricultural production is reliant on rainfall in the South African context where the majority of crops are rainfed (Hartley et al., 2021) . This dependence is further heightened by the country's relatively limited water resources (Government of South Africa, 2024). Hence, non-extreme positive precipitation shocks may aid agricultural production. This may, in turn, yield favourable outcomes in the agricultural sector as well as upstream and downstream sectors and ultimately allow for a reduction in $f(D_t/E_t)$ and ϕ_t.

While we present an array of explanations for how climate shocks interact with monetary policy transmission, we ultimately focus on whether the overall responsiveness of credit conditions to monetary policy shocks changes in the presence of climate shocks, without attributing this change to a specific household, corporate or banking-sector mechanism.

3 Data and methodology

3.1 Data

The analysis draws on three main data sources: climate shocks, monetary policy shocks, and credit extension data. The sample covers South Africa over the period 2011–2025. All series are observed at a quarterly frequency, determined by the availability of the monetary policy shock data; higher-frequency series are aggregated to quarterly values by taking averages or sums where appropriate.

Climate data: Temperature and precipitation data are sourced from the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts[1]. ERA5 provides globally gridded, hourly atmospheric estimates at a 31 km horizontal resolution. Grid-level values are weighted by population and aggregated to monthly averages over South Africa's land area, from which we construct climate shock variables that capture deviations from long-run historical norms in temperature and precipitation.

The climate shocks are calculated as follows:

$$ClimateShock_t = \frac{T_t - \overline{T}_t}{\sigma(T_t)}, \tag{3}$$

where $ClimateShock_t$ is the temperature or precipitation shock at time t and T_t is the observed population-weighted climate variable. For both temperature and precipitation, $\overline{T}_t$ is the calendar-month mean computed over the full sample period (e.g., the mean of all January values from 2011 to 2025), and $\sigma(T_t)$ is the corresponding within-month standard deviation. The resulting shock series are shown in Figure A3 in the Appendix.

Monetary policy data: The monetary policy shock series is constructed using high-frequency policy surprises for South Africa from Pirozhkova and Viegi (2025), alongside the repurchase (repo) rate from the South African Reserve Bank (SARB). The high-frequency surprises serve as an instrument to purge the endogenous component of policy rate changes, yielding a shock series that satisfies the exogeneity requirement of the regression framework de-

[1] see https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview

scribed in Section 3.2. For robustness, we consider both market-based shocks from Pirozhkova and Viegi (2025) (see Figure A4) and a theory-based shock derived from the policy rate, as described in Section 3.2 below (see Figure A5). Surprises are timed to coincide with policy rate announcement dates and aggregated to a quarterly frequency. Underlying the calculation of the are the forecasts of the real GDP growth, the policy rate (repo-rate), and the inflation rate from Bloomberg Consensus.

Credit extension data: Private sector credit extension data — disaggregated by borrower type (households and corporates) at the bank level — are obtained from the SARB's BA900 and BA930 bank compliance reports.[2] These series include both credit volumes and lending rates, and are constructed following Sibande et al. (2025) and Sibande and Milne (2025) by sorting and aggregating individual bank records into the broad credit categories used in the study: unsecured, secured, and mortgage credit. We use the log of credit volumes and make no other transformations to the other credit data. Summary charts are provided in Figure A1 and Figure A2.

3.2 Methodology

3.2.1 Shock identification

Rather than using the policy interest rate directly — which responds to prevailing economic conditions and is therefore endogenous — we use an unexpected change in monetary policy, commonly called a monetary policy shock, to isolate the causal effect of policy on credit markets (Miyajima, 2024). We construct this shock following Merrino (2022), who adapts the approach of Romer and Romer (2004) to the South African context. The idea is to regress the change in the repo rate forecast on variables that policymakers were already tracking, and treat whatever is left over — the part of the rate change that cannot be explained by those forecasts — as the true policy surprise. Formally:

$$\Delta r_t = \alpha + \tilde{r_t} + \sum_{t=0}^{2} \beta_t \Delta \tilde{y_t} + \sum_{t=0}^{2} \delta_t \tilde{\pi_t} + \varepsilon_t, \tag{4}$$

[2]BA930 data at the bank level are not publicly available.

where Δr_t is the change in the repo rate forecast, $\tilde{r_t}$ is the forecasted repo rate, $\tilde{y_t}$ is the forecasted real GDP growth rate, and $\tilde{\pi_t}$ is the forecasted inflation rate. The residual ε_t — the unexplained part of the forecast revision — is our monetary policy shock, MPShock_t.[3]

As an alternative, we construct a second measure of the monetary policy surprise following Miyajima (2024). This approach uses forecast errors — the gap between what forecasters expected and what actually occurred — rather than SARB forecasts. The construction proceeds in two steps. First, for each variable i, the forecast error is:

$$FE_t^i = EXP_t^i - ACT_t^i, \tag{5}$$

where $i \in \{\text{policy rate}, \text{inflation}, \text{GDP growth}\}$, ACT_t^i is what was actually observed, and EXP_t^i is the Bloomberg consensus forecast made beforehand. In the second step, the policy rate forecast error is regressed on the inflation and output growth forecast errors:

$$FE_t^r = \alpha + \beta FE_t^\pi + \gamma FE_t^y + \epsilon_t, \tag{6}$$

where FE_t^π is the inflation forecast error and FE_t^y is the GDP growth forecast error. The residual ϵ_t is the monetary policy surprise — the part of the policy rate miss that has nothing to do with inflation or growth news, and is therefore not anticipated by market participants.

As a further robustness check, we use market-based monetary policy surprises built from a high-frequency identification approach (Gürkaynak et al., 2005). The core idea is to measure how financial market prices move in a narrow window around each MPC announcement: since little else happens in that window, any price change can be attributed to the policy news itself (Nakamura and Steinsson, 2018). The target monetary policy surprise, which is related to the market reaction immediately after an announcement is taken from Pirozhkova and Viegi (2025). Each component is used in turn as an alternative measure of MPShock_t in Equation 8 and Equation 9.

** conceptually, what are the differences, how do we think these link to the credit market

[3] We include forecasts up to two quarters ahead to account for the common assumption among forecasters that policy is unchanged in the very near term. See Romer and Romer (2004) for details.

products in turn. **

As policymakers are aware of climate conditions and may already adjust rates in response to them. If so, our monetary policy shocks would partly reflect climate events rather than pure policy decisions, which would confound our estimates. To address this, we regress each monetary policy shock on the climate shock measures:

$$\text{MPShock}_t^i = \text{ClimateShock}_t^i + \varepsilon_t^i, \tag{7}$$

where MPShock_t^i is each of the monetary policy shocks described above and ClimateShock_t^i are the temperature and precipitation shocks. The residual ε_t^i — the part of the policy shock that cannot be explained by climate conditions — is retained as a "purged" shock. We use these purged shocks in all subsequent estimations. ** To find sources**

3.2.2 Panel estimation

Our first estimating equation tests whether the effect of a monetary policy shock on credit is amplified or dampened when a climate shock occurs at the same time:

$$\begin{aligned}\text{Credit}_{s,t} =& \beta_0 + \beta_1 \text{ClimateShock}_t^k + \beta_2 \text{MPShock}_t + \beta_3 (\text{ClimateShock}_t^k \times \text{MPShock}_t) + \\ & \Gamma X_t + \gamma_s + \tau t + \varepsilon_{s,t},\end{aligned} \tag{8}$$

where $\text{Credit}_{s,t}$ is credit growth for agent s (households or corporates) at time t, and ClimateShock_t^k captures climate shocks — temperature, precipitation, and extreme wet and dry events, indexed by k. The interaction term $\text{ClimateShock}_t^k \times \text{MPShock}_t$ is the key coefficient of interest: a significant estimate would indicate that climate shocks alter how monetary policy transmits through credit markets. γ_s are agent fixed effects (absorbing time-invariant differences across borrower types), τt is a linear time trend, and $\varepsilon_{s,t}$ is the error term. X_t is a vector of controls added as a robustness check, including GDP growth, the USD/ZAR exchange rate, and the unemployment rate.

The panel regression above captures the contemporaneous relationship but cannot tell us

how long any effect lasts. To trace the full dynamic response of credit over time, we use the local projections method of Jordà (2005), estimating a separate regression for each horizon $h = 0, 1, ..., H$:

$$\begin{aligned} Credit_{s,t+h} - Credit_{s,t-1} = &\beta_0^h + \beta_1^h \text{ClimateShock}_t^k + \beta_2^h \text{MPShock}_t + \\ &\beta_3^h (\text{ClimateShock}_t^k \times \text{MPShock}_t) + \Gamma X_t + \gamma_s + \tau t + \varepsilon_{s,t}^h, \end{aligned} \tag{9}$$

where $\text{Credit}_{s,t+h} - \text{Credit}_{s,t-1}$ is the cumulative change in credit growth from period $t-1$ to $t+h$, and h is the horizon in quarters. Plotting the estimated $\widehat{\beta^h}_2$, and $\widehat{\beta^h}_2 + \widehat{\beta^h}_3$ across horizons gives the impulse response functions, showing the differential impact of climate on credit in the quarters following a shock. All regressions use Driscoll-Kraay standard errors to account for correlation across agents and over time (Driscoll and Kraay, 1998).

4 Results

4.1 Contemporaneous Evidence: Does Climate Alter the Credit Response to Monetary Policy?

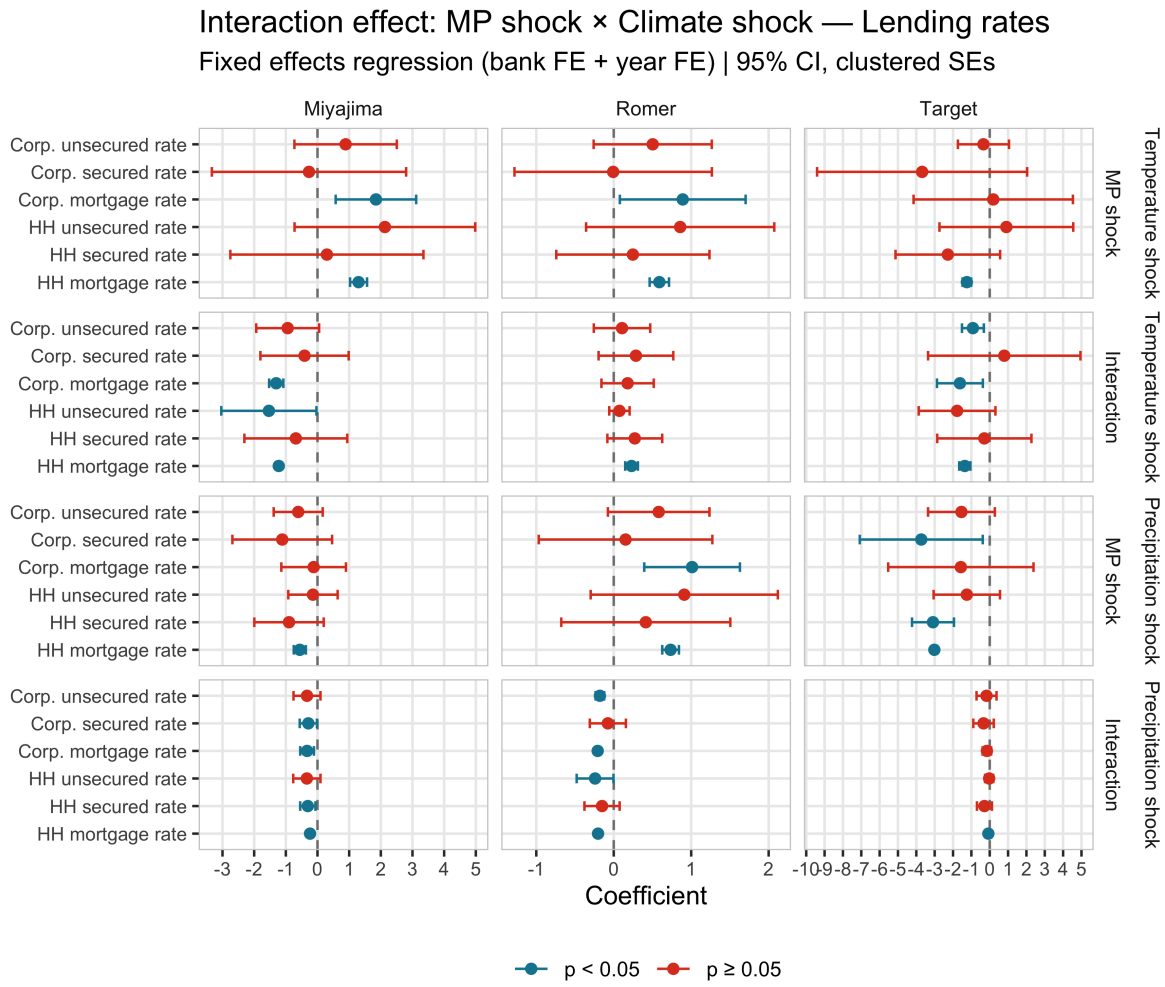

Figure 1: Interaction between climate and monetary policy shocks on lending rates

Note: Each point shows the estimated coefficients $\hat{\beta}_2$ on the MPShock and $\hat{\beta}_3$ on the interaction term ClimateShock × MPShock from Equation 8, with 95% confidence intervals. Teal points indicate statistical significance at the 5% level; red points are insignificant. Rows correspond to individual lending rate series disaggregated by borrower type and product; columns correspond to the three monetary policy surprise measures. All models include bank and quarter-of-year fixed effects with standard errors clustered at the bank level.

4.2 The Dynamic Credit Channel: How Climate–Monetary Policy Interactions Evolve Over Time

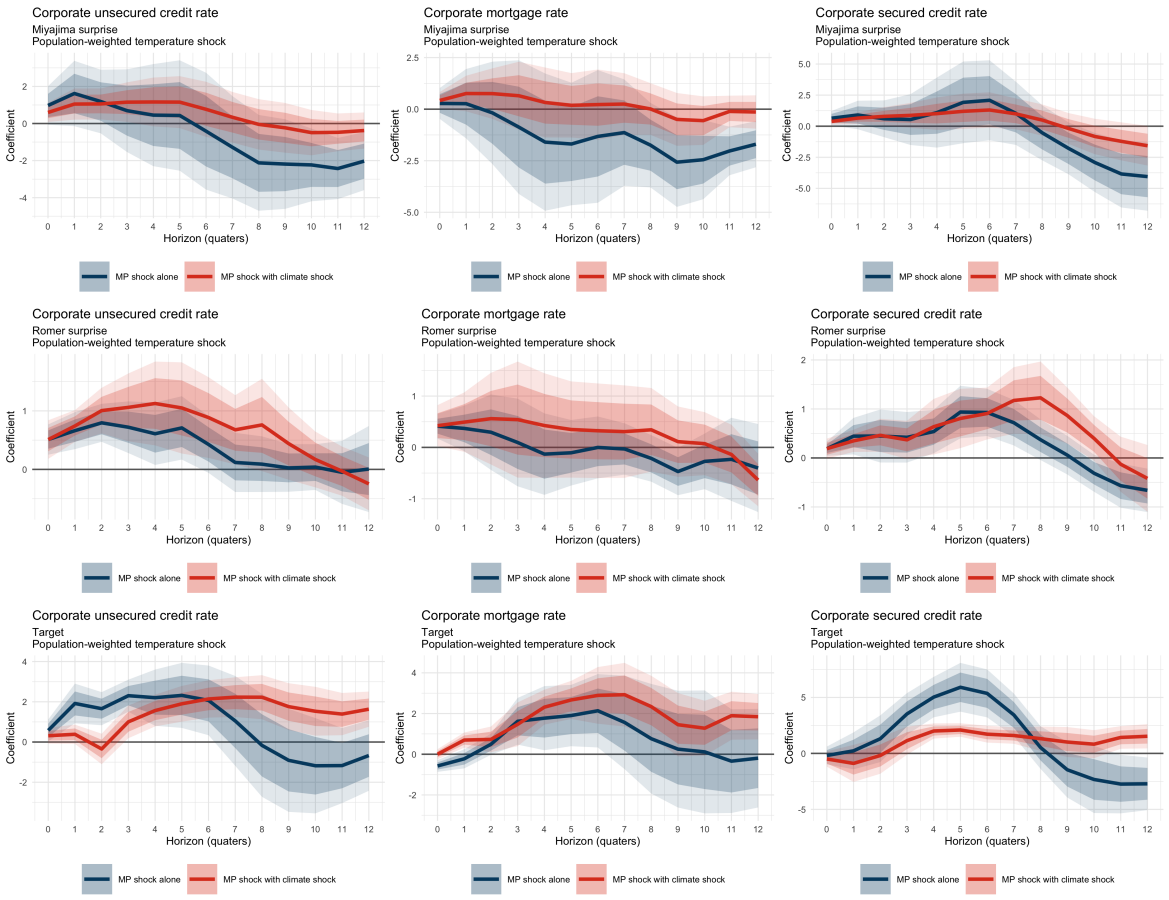

Figure 2: Impulse responses of corporate lending rates to monetary policy shocks: temperature shock interaction

Note: Impulse response functions (IRFs) from panel local projections estimated at horizons $h = 0, \ldots, 12$ quarters following Equation 9. The blue line shows the cumulative response of the corporate lending rate to the monetary policy shock alone ($\hat{\beta}_2$); the red line shows the response when the same shock coincides with the population-weighted temperature shock ($\hat{\beta}_2 + \hat{\beta}_3$). Shaded bands are 68% and 90% Driscoll-Kraay confidence intervals. Rows correspond to corporate unsecured, secured, and mortgage lending rates; columns correspond to the Miyajima, Romer, and Target monetary policy surprise measures.

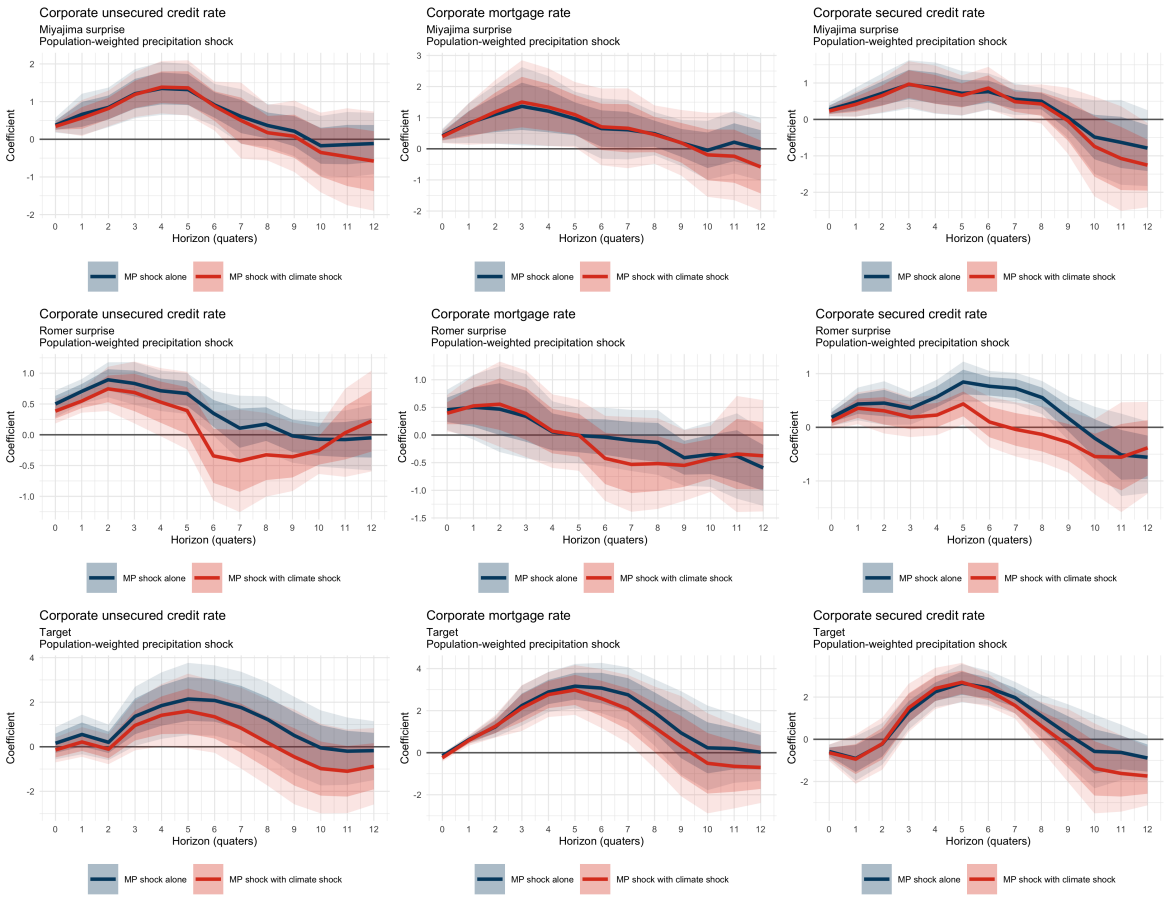

Figure 3: Impulse responses of corporate lending rates to monetary policy shocks: precipitation shock interaction

Note: As Figure 2 but for the population-weighted precipitation shock. The red line shows the corporate lending rate response when the monetary policy shock coincides with a precipitation shock ($\hat{\beta}_2 + \hat{\beta}_3$).

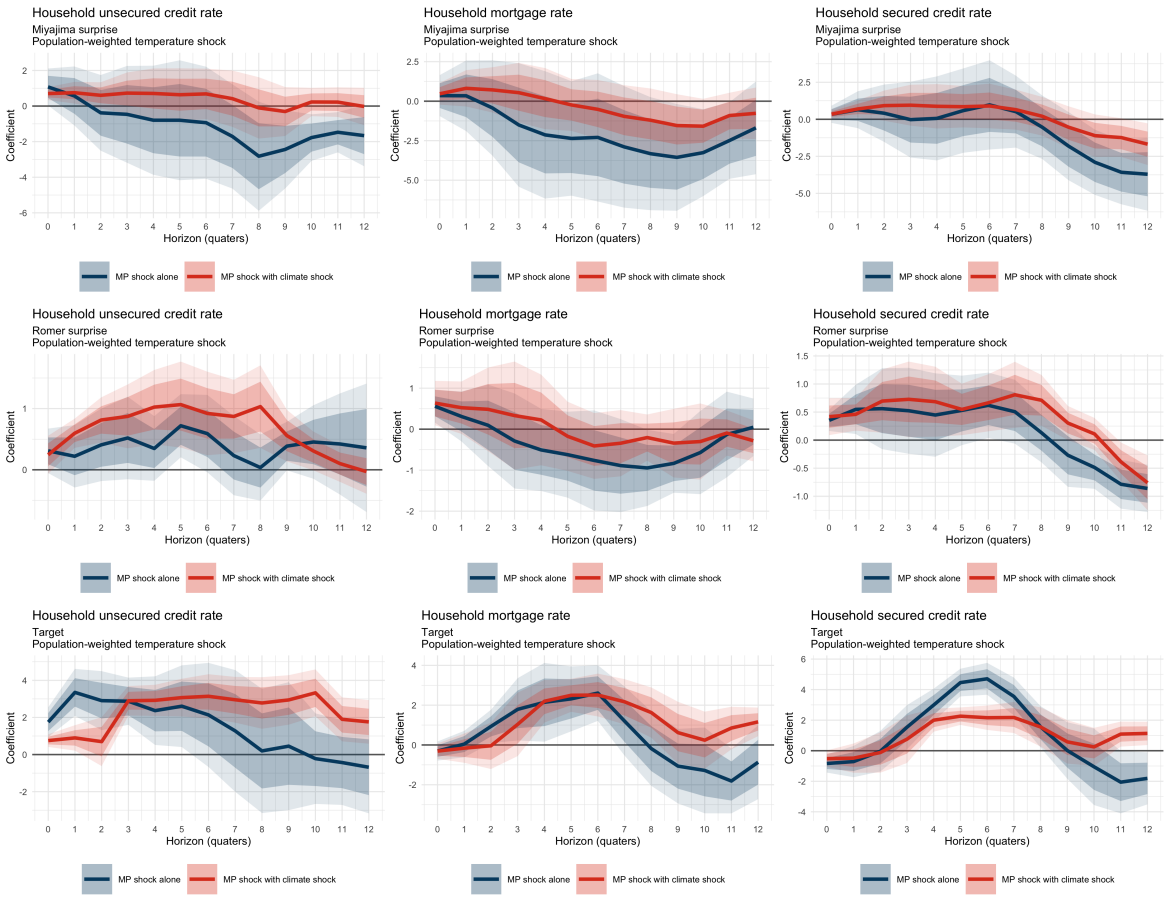

Figure 4: Impulse responses of household lending rates to monetary policy shocks: temperature shock interaction

Note: As Figure 2 but for household lending rates (unsecured, secured, and mortgage). The gap between the blue and red lines at each horizon represents the climate amplification or dampening of monetary policy transmission.

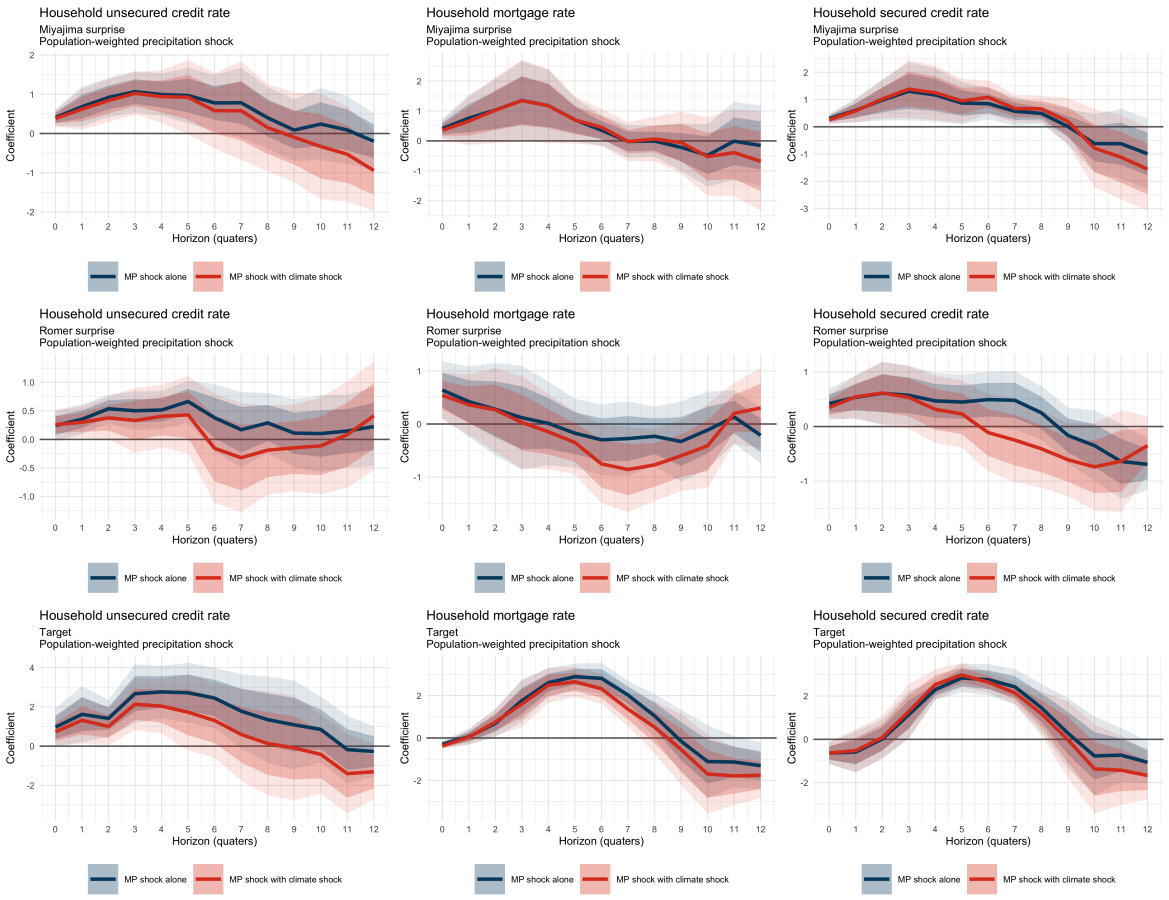

Figure 5: Impulse responses of household lending rates to monetary policy shocks: precipitation shock interaction

Note: As Figure 4 but for the population-weighted precipitation shock. Rows correspond to household unsecured, secured, and mortgage lending rates.

4.3 The Direction of the Climate Credit Channel: Amplification, Dampening, and Product Heterogeneity

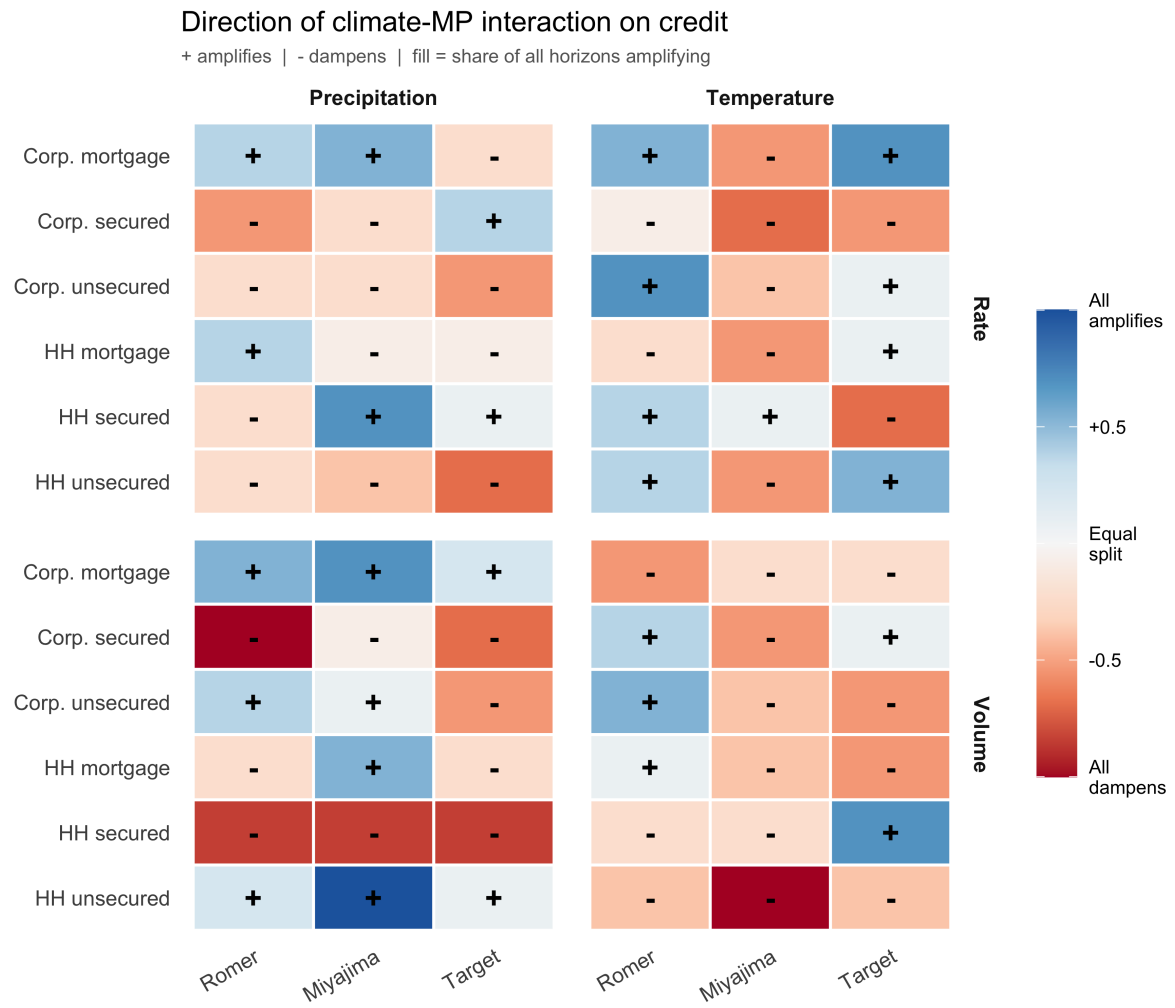

Figure 6: Direction and magnitude of the climate–monetary policy interaction across credit products and shock measures

Note: Heatmap summarising the direction and strength of climate shock effects on monetary policy transmission, derived from panel local projections estimated at horizons $h = 0, \ldots, 12$ quarters. Direction is determined by comparing the absolute magnitudes of the two IRF paths at each horizon: a horizon is classified as *amplifying* when $|\hat{\beta}_2 + \hat{\beta}_3| > |\hat{\beta}_2|$ (the credit response is larger in absolute value when the climate shock coincides with the monetary policy shock) and *dampening* otherwise. The symbol "+" indicates that the majority of horizons show amplification; "−" indicates the majority show dampening. Colour intensity reflects the *amplification score*, defined as $2 \times \bar{p} - 1$, where $\bar{p}$ is the proportion of horizons classified as amplifying: deep blue ($\approx +1$) indicates near-universal amplification, deep red (≈ -1) indicates near-universal dampening, and white (≈ 0) indicates an approximately equal split. Rows correspond to individual credit products disaggregated by borrower type; columns correspond to three monetary policy surprise measures (Romer, Miyajima, and Target factor). Panels separate the results by credit type (rates vs. log volumes) and climate shock (temperature vs. precipitation).

4.4 Persistence and Timing of the Climate Credit Channel

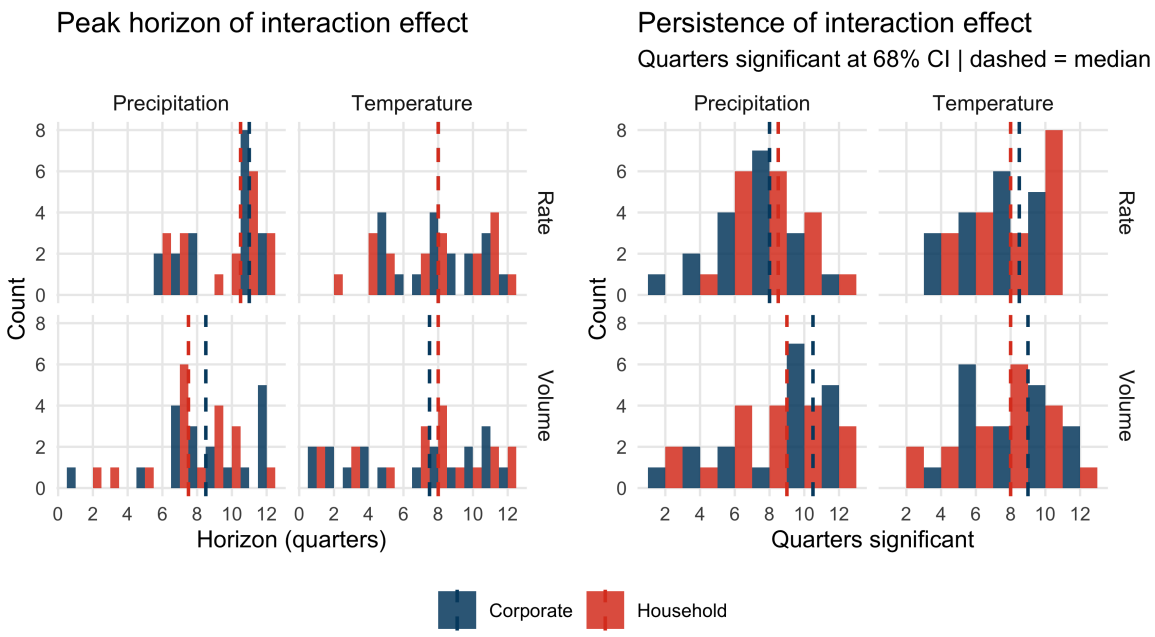

Figure 7: Timing and persistence of the climate–monetary policy interaction

Note: Distribution of two timing statistics computed from the local projection interaction coefficients. *Left panel:* the horizon at which the absolute interaction effect reaches its peak, across all credit product and monetary policy shock combinations. *Right panel:* the number of quarters for which the interaction coefficient is statistically significant at the 68% confidence level (a measure of persistence). Dashed lines mark the median for each borrower group. Results are disaggregated by credit type (rates vs. log volumes) and climate shock (temperature vs. precipitation).

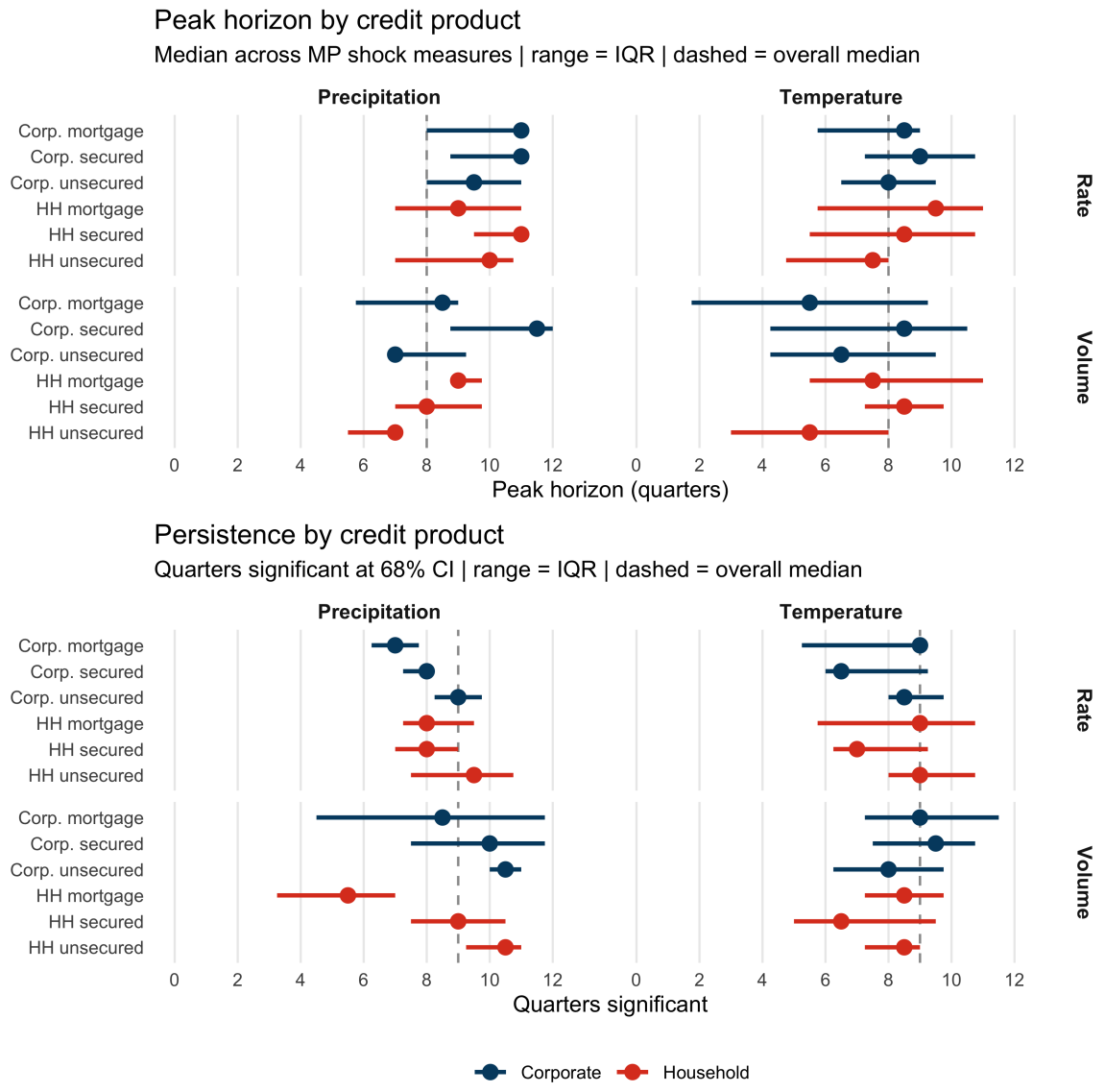

Figure 8: Timing and persistence of the climate–monetary policy interaction by credit product

Note: As Figure 7 but disaggregated by individual credit product rather than pooled across products. Points show the median peak horizon (left panel) and median quarters significant (right panel) across all three monetary policy surprise measures, with interquartile range bars. The dashed line marks the overall median across all products and shock combinations. Corporate products are shown in dark blue; household products in red.

4.5 When Climate Shocks Are Severe: Asymmetric Effects on Credit Transmission

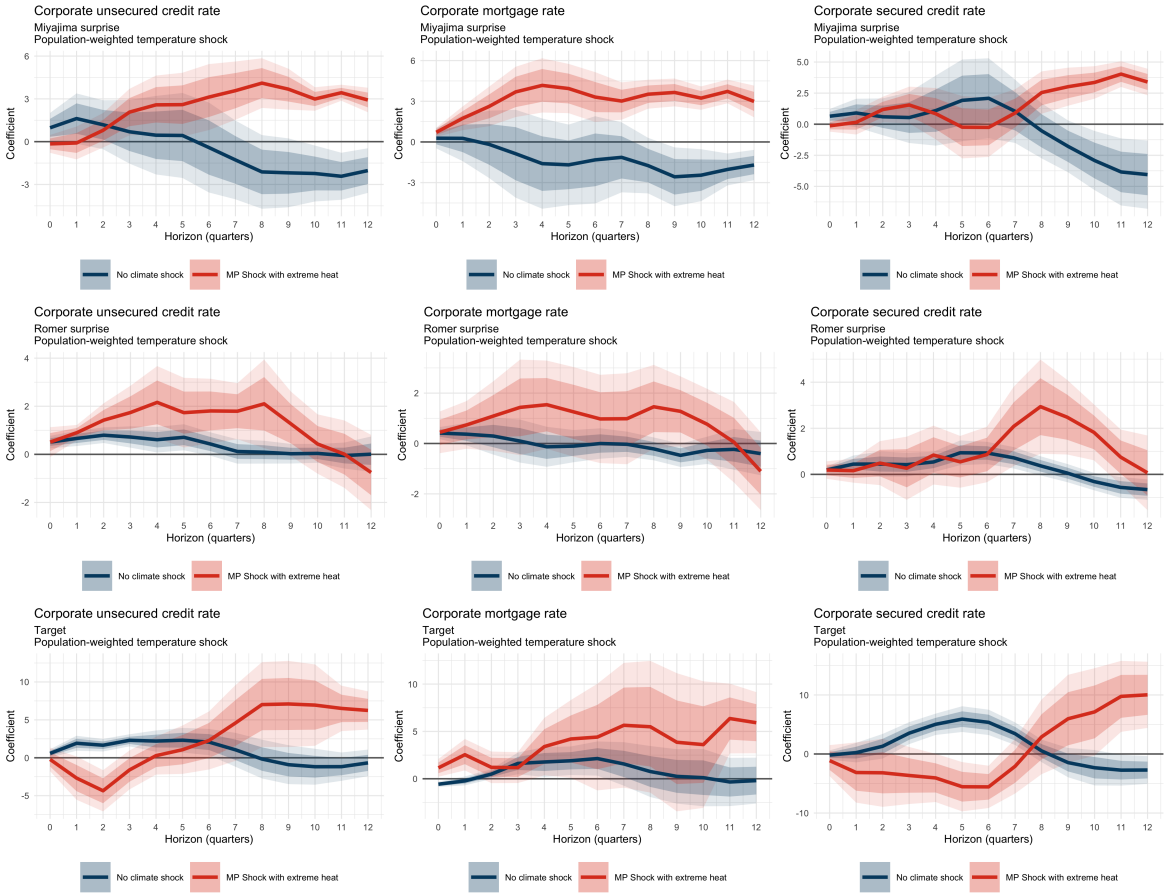

Figure 9: Asymmetric impulse responses of corporate lending rates under extreme temperature states

Note: Impulse response functions from asymmetric panel local projections. The blue line shows the cumulative response of corporate lending rates to the monetary policy shock in the absence of a climate shock; the red line shows the response evaluated at an extreme positive temperature state (+3 SD). The gap between the two paths is the marginal climate amplification or dampening effect at that climate intensity. Shaded bands are 68% and 90% Driscoll-Kraay confidence intervals. Rows correspond to corporate unsecured, secured, and mortgage lending rates.

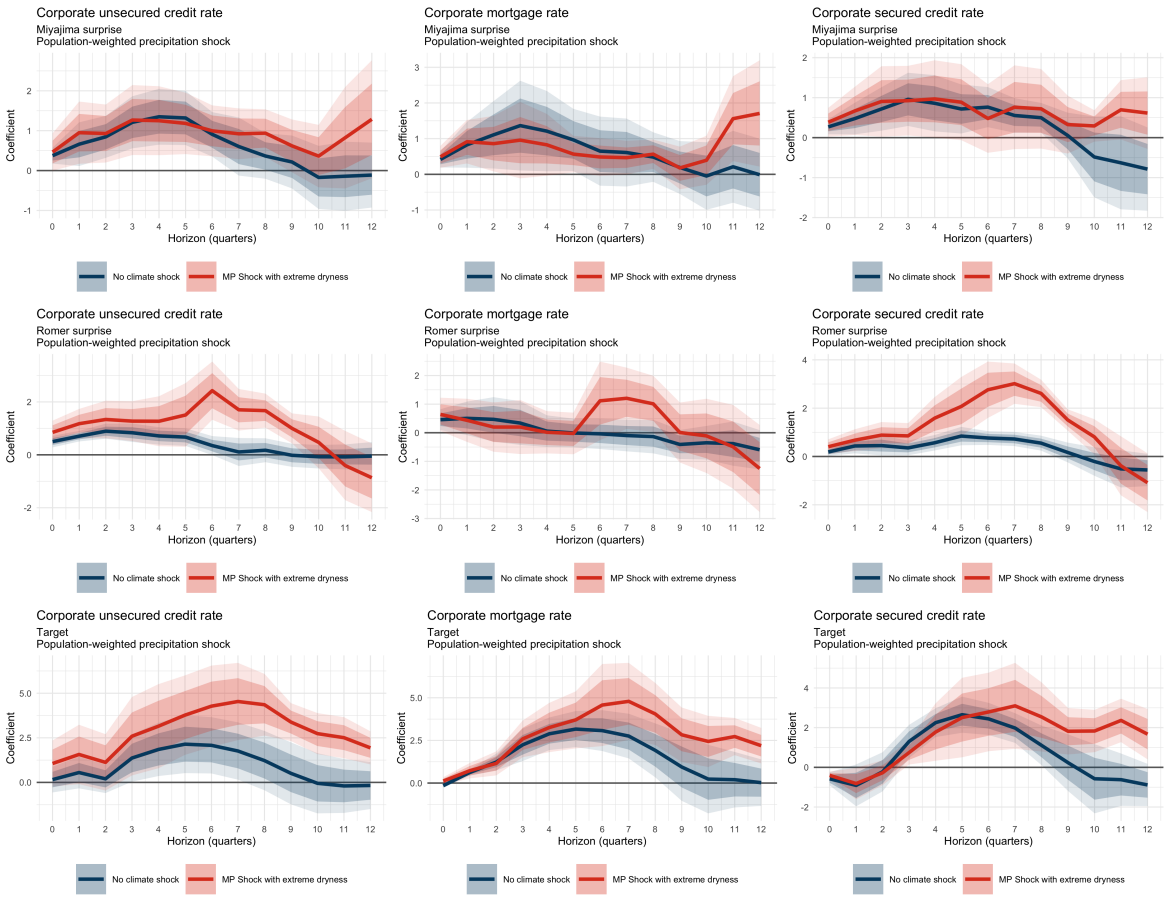

Figure 10: Asymmetric impulse responses of corporate lending rates under extreme precipitation states

Note: As Figure 9 but evaluated at an extreme negative precipitation state (−3 SD, i.e. drought conditions). A gap between the red and blue lines indicates that drought amplifies or dampens the transmission of monetary policy to corporate lending rates relative to the no-climate-shock baseline.

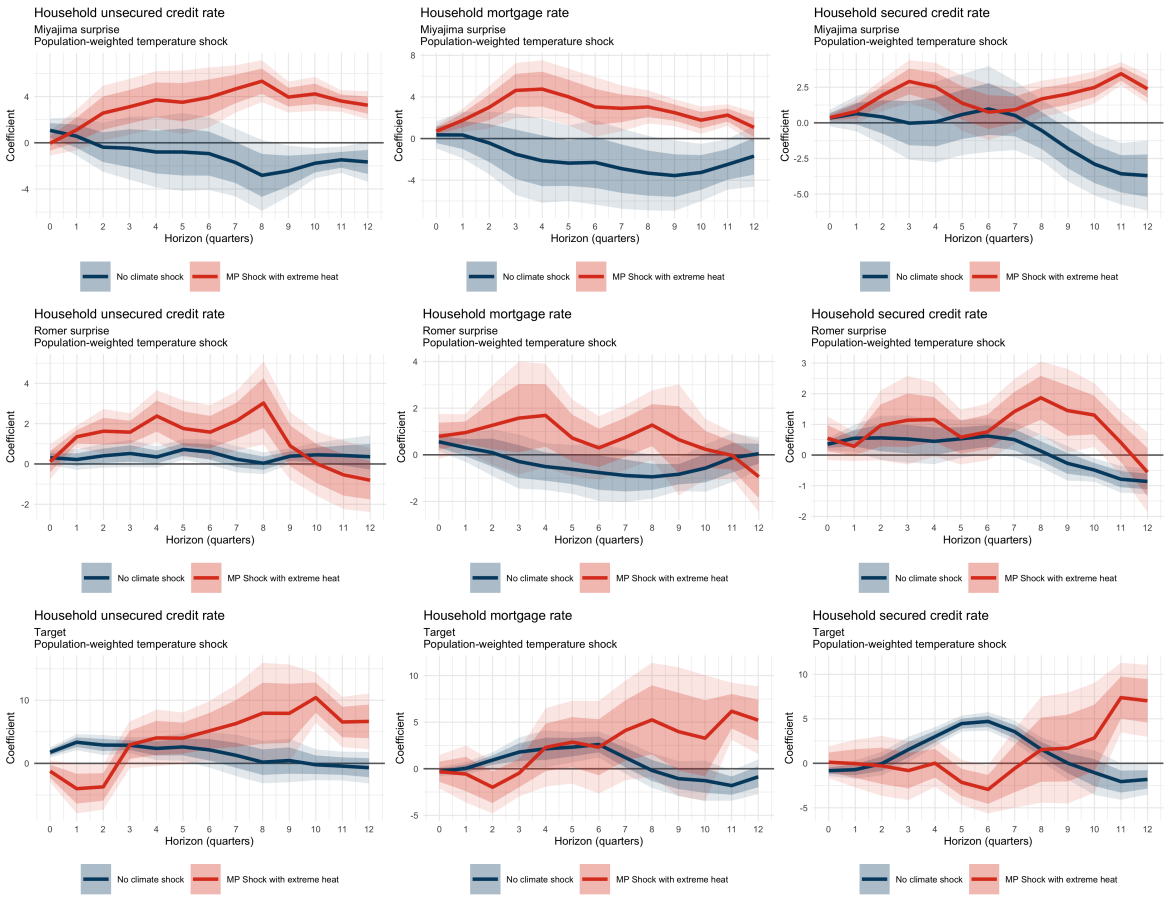

Figure 11: Asymmetric impulse responses of household lending rates under extreme temperature states

Note: As Figure 9 but for household lending rates (unsecured, secured, and mortgage). Compares the monetary policy transmission response in the absence of a climate shock against the response under an extreme positive temperature state (+3 SD).

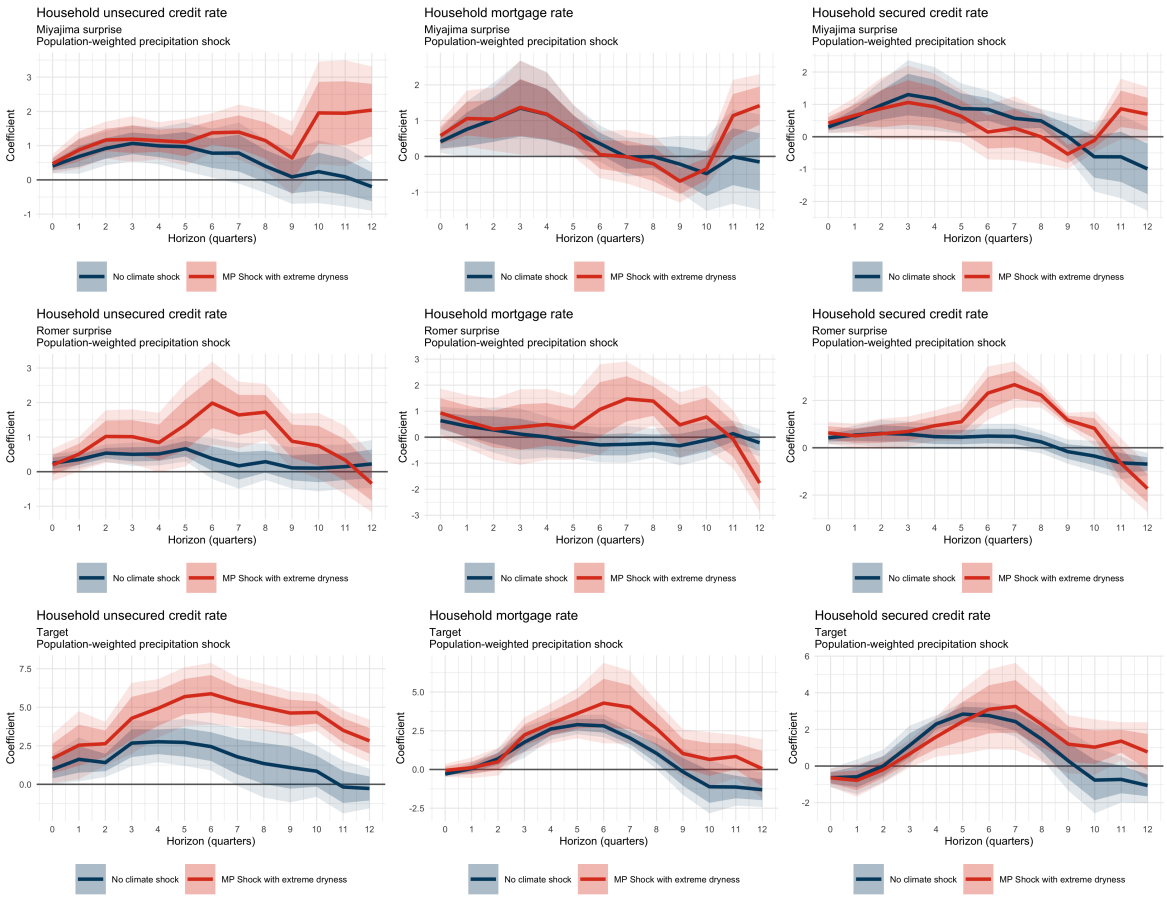

Figure 12: Asymmetric impulse responses of household lending rates under extreme precipitation states

Note: As Figure 11 but evaluated at an extreme negative precipitation state (−3 SD). Rows correspond to household unsecured, secured, and mortgage lending rates.

4.6 Discussion

5 Conclusion

This paper investigates whether physical climate shocks alter the transmission of monetary policy through the credit channel in South Africa. Using bank-level panel data on credit extension and lending rates over 2011–2025, we estimate the interaction between three monetary policy surprise measures and two population-weighted climate shocks — temperature and pre-

cipitation — within both a panel fixed effects framework and a local projections framework. The evidence points to three broad conclusions.

Climate shocks affect monetary policy transmission primarily through lending rates, not credit volumes. The contemporaneous fixed effects estimates show that significant interactions between climate and monetary policy shocks are concentrated in lending rates across both corporate and household borrowers, while log credit volumes yield little consistent evidence of interaction. This asymmetry is consistent with a risk-repricing mechanism: banks adjust the price of credit in response to climate-related deterioration in borrower balance sheets — captured by changes in the external finance premium — rather than rationing quantities outright. The implications for the balance sheet channel are therefore mediated through the cost of borrowing rather than its availability, at least at the level of aggregation available here.

The clearest and most persistent results arise under extreme climate conditions. While the average-shock local projections reveal significant interactions at the 68% confidence level across a large share of horizon-product combinations, the direction and magnitude of the interaction is heterogeneous and, for some combinations, fragile. The asymmetric local projections — evaluated at ± 3 standard deviation climate states — produce sharper and more durable results. Under extreme temperature conditions in particular, the IRF paths for corporate and household lending rates diverge substantially from the no-climate-shock baseline and remain separated well beyond the median peak horizon of 8–10 quarters. This suggests that the credit channel implications of climate shocks are primarily a tail-risk phenomenon: it is severe and unusual climate episodes that create the most consequential frictions for monetary policy transmission, rather than moderate seasonal variability.

The interaction can amplify or dampen monetary policy transmission depending on the type of shock. Rather than a uniform dampening of the credit channel — as a simple external-finance-premium argument might predict — we find that whether a climate shock amplifies or dampens monetary policy depends jointly on the climate shock measure, the monetary policy surprise measure, and the credit product. This heterogeneity is economically meaningful: temperature and precipitation shocks affect different borrower populations through different sectoral exposures, The interaction of these two sources of heterogeneity produces the mixed

amplification and dampening pattern documented in the heatmap analysis.

These findings carry implications for monetary policy practice. Climate shocks share features with traditional supply shocks — they alter production costs, damage capital, and compress firm profitability — but their interaction with financial frictions gives them a distinctly financial-stability dimension. In particular, the evidence that severe climate shocks can make policy more restrictive through the credit channel than intended suggests that central banks operating in climate-vulnerable economies may benefit from incorporating physical climate risk into their models of credit transmission. Ignoring the coincidence of climate and monetary policy shocks risks misjudging the stance of policy, particularly at the extremes of the climate distribution.

Several limitations bound these conclusions and point to directions for future research. First, the absence of bank-level sectoral credit data prevents us from tracing the interaction to specific climate-exposed industries — agricultural, energy, and construction lending may behave differently from the aggregate — and obtaining sector-level exposure data would substantially deepen the interpretation of the heterogeneity we document. Second, quarterly data limits our ability to identify the precise timing of the credit response relative to the shock; higher-frequency data would sharpen estimates of the impact horizon and the speed of rate adjustment. Third, the monetary policy shock measures used here are constructed from standard Romer-Romer and high-frequency identification approaches; less conventional measures that more directly capture the SARB's climate-related policy communications, or that separate the physical from the financial-stability component of climate news, may reveal additional dimensions of the interaction not captured in the present framework.

References

Aglietta, M. and Espagne, É. (2016). *Climate and Finance Systemic Risks, More than an Analogy?: The Climate Fragility Hypothesis*. CEPII, Centre d'etudes prospectives et d'informations internationales.

Altavilla, C., Canova, F., and Ciccarelli, M. (2020). Mending the broken link: Heterogeneous bank lending rates and monetary policy pass-through. *Journal of Monetary Economics*, 110:81–98.

Altavilla, C., Gürkaynak, R. S., and Quaedvlieg, R. (2024). Macro and micro of external finance premium and monetary policy transmission. *Journal of Monetary Economics*, 147:103634.

Ashcraft, A. B. and Campello, M. (2007). Firm balance sheets and monetary policy transmission. *Journal of Monetary Economics*, 54(6):1515–1528.

Aysun, U. and Hepp, R. (2013). Identifying the balance sheet and the lending channels of monetary transmission: A loan-level analysis. *Journal of banking & Finance*, 37(8):2812–2822.

Bean, C., Larsen, J. D. J., and Nikolov, K. (2002). Financial frictions and the monetary transmission mechanism: theory, evidence and policy implications. ECB Working Paper 113, Frankfurt a. M.

Bernanke, B. S. and Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic perspectives*, 9(4):27–48.

Dafermos, Y., Nikolaidi, M., and Galanis, G. (2018). Climate change, financial stability and monetary policy. *Ecological Economics*, 152:219–234.

Driscoll, J. C. and Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics*, 80(4):549–560.

Duanmu, J., Li, Y., Lin, M., and Tahsin, S. (2022). Natural disaster risk and residential mortgage lending standards. *Journal of Real Estate Research*, 44(1):106–130.

Gallagher, J. and Hartley, D. (2017). Household finance after a natural disaster: The case of hurricane katrina. *American Economic Journal: Economic Policy*, 9(3):199–228.

Government of South Africa (2024). Water and sanitation. About South Africa. Source: Official Guide to South Africa 2023/24.

Gürkaynak, R. S., Sack, B., and Swanson, E. (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.

Hall, S. (2001). Credit channel effects in the monetary transmission mechanism. *Bank of England. Quarterly Bulletin*, 41(4):442.

Hartley, F., Gabriel, S., Cullis, J., and Arndt, C. (2021). Climate uncertainty and agricultural vulnerability in south africa. Policy Brief 02/21, Southern Africa – Towards Inclusive Economic Development (SA-TIED), UNU-WIDER.

Ho, A. T., Huynh, K. P., Jacho-Chávez, D. T., and Vallée, G. (2023). We didn't start the fire: Effects of a natural disaster on consumers' financial distress. *Journal of Environmental Economics and Management*, 119:102790.

Jordà, O. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

Kleimeier, S. and Sander, H. (2006). Expected versus unexpected monetary policy impulses and interest rate pass-through in euro-zone retail banking markets. *Journal of Banking & Finance*, 30(7):1839–1870. Special Section: Banking and Finance in an Integrating Europe.

Laumer, S. and Schaffer, M. (2025). Monetary policy transmission under supply chain pressure. *European Economic Review*, 172:104949.

Merrino, S. (2022). Monetary policy and wage inequality in South Africa. *Emerging Markets Review*, 53:100911.

Miyajima, K. (2024). Monetary policy, inflation and distributional impact: South Africa's case. *South African Journal of Economics*, 92(3):305–330.

Nakamura, E. and Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: The information effect. *Quarterly Journal of Economics*, 133(3):1283–1330.

Pirozhkova, E. and Viegi, N. (2025). The bank lending channel of monetary policy transmission in South Africa. *The Quarterly Review of Economics and Finance*, page 102041.

Romer, C. D. and Romer, D. H. (2004). A new measure of monetary shocks: Derivation and implications. *American economic review*, 94(4):1055–1084.

Sharpe, S. A. and Suarez, G. A. (2020). Why isn't business investment more sensitive to interest rates? evidence from surveys. *Management Science*, 67(2):720–741.

Sibande, X. and Milne, A. (2025). The Impact of Basel III Implementation on Bank Lending in South Africa. *South African Journal of Economics*, 93(1):6–26.

Sibande, X., Nxumalo, D., Mncube, K., Koch, S., and Viegi, N. (2025). Regulation and Bank Lending in South Africa: A Narrative Index Approach. *South African Journal of Economics*, 93(1):73–85.

Slacalek, J., Tristani, O., and Violante, G. L. (2020). Household balance sheet channels of monetary policy: A back of the envelope calculation for the euro area. *Journal of Economic Dynamics and Control*, 115:103879. St. Louis Fed -JEDC-SCG-SNB-UniBern Conference, titled "Disaggregate Data and Macroeconomic Models".

Xu, J., Cai, D., and Zhu, J. (2025). Navigating the green wave: Urban climate adaptation and firms' investment decisions-evidence from china. *Energy Economics*, 141:108087.

A Appendix

A.1 Data

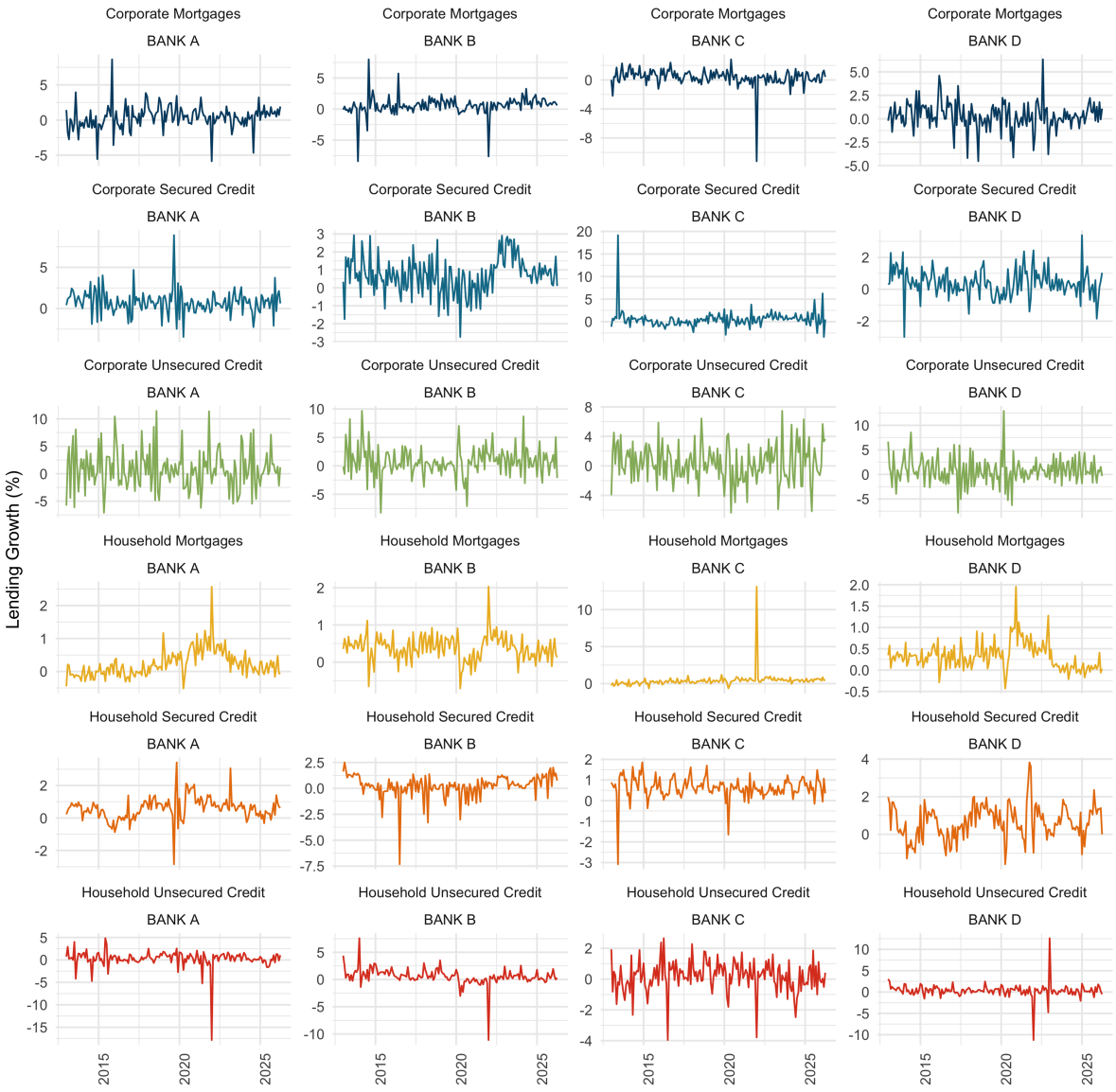

Figure A1: Private sector credit extension by borrower type and product, 2011–2025

Note: Log of quarterly credit extension at the bank level, disaggregated by borrower type (corporate and household) and product category (unsecured, secured, and mortgage credit). Series are constructed from the South African Reserve Bank's BA900 and BA930 bank compliance reports following Sibande et al. (2025) and Sibande and Milne (2025). The sample covers the period 2011Q1–2025Q1 and is restricted to the major South African banking groups.

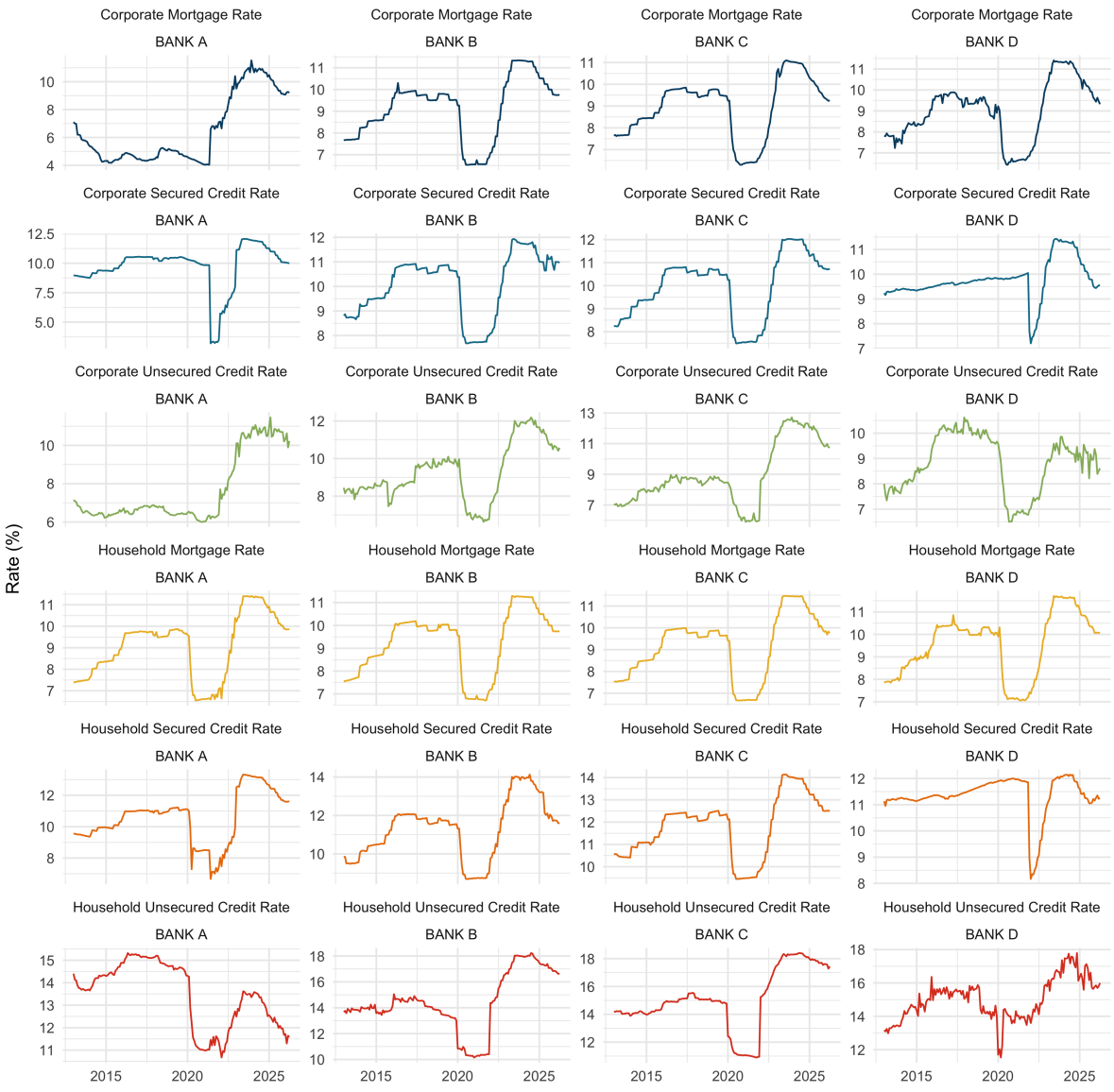

Figure A2: Bank-level lending rates by borrower type and credit product, 2011–2025

Note: Quarterly lending rates at the bank level, disaggregated by borrower type and product (unsecured, secured, and mortgage credit for both corporates and households). Rates are expressed in percentage points and are sourced from the SARB BA930 compliance reports. Unlike credit volumes, lending rates are not log-transformed. The pronounced heterogeneity across banks and products reflects differences in risk pricing, collateral requirements, and borrower characteristics.

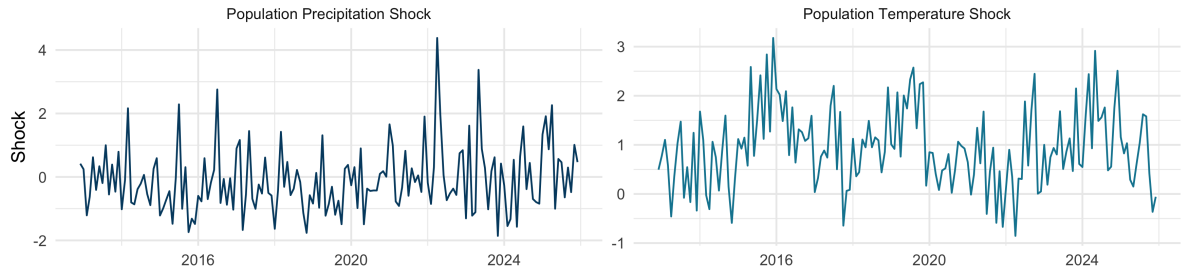

Figure A3: Population-weighted temperature and precipitation shocks for South Africa, 2011–2025

Note: Standardised climate shock series constructed from ERA5 reanalysis data (European Centre for Medium-Range Weather Forecasts) at a 31 km horizontal resolution. Each observation is the deviation of the population-weighted monthly average from the long-run calendar-month mean, expressed in standard deviation units: $\text{ClimateShock}_t = (T_t - \bar{T}_t)/\sigma(T_t)$. Positive values indicate warmer or wetter conditions than the historical norm; negative values indicate cooler or drier conditions. Grid-level values are weighted by population density and aggregated to a quarterly frequency by averaging within-quarter monthly values.

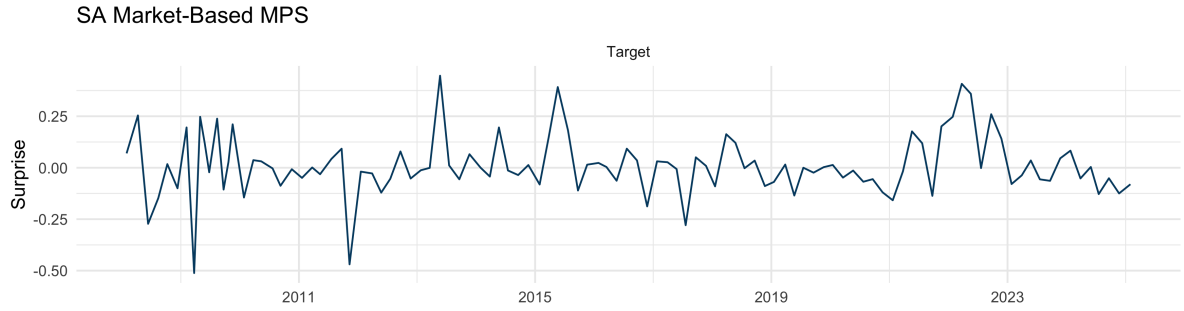

Figure A4: High-frequency market-based monetary policy surprise factors, 2011–2025

Note: Monetary policy surprise factors derived from the high-frequency identification approach of Gürkaynak et al. (2005) and constructed for South Africa by Pirozhkova and Viegi (2025). Each factor is extracted from movements in financial market prices in a narrow window around Monetary Policy Committee (MPC) announcement dates. Four components are shown: the *target factor* (surprise in the level of the policy rate), the *forward guidance factor* (surprise in the expected future path of policy), the *central bank information factor* (economic outlook news revealed by the central bank's communication), and the *country risk factor* (shifts in sovereign risk perceptions around MPC dates). Series are aggregated to a quarterly frequency.

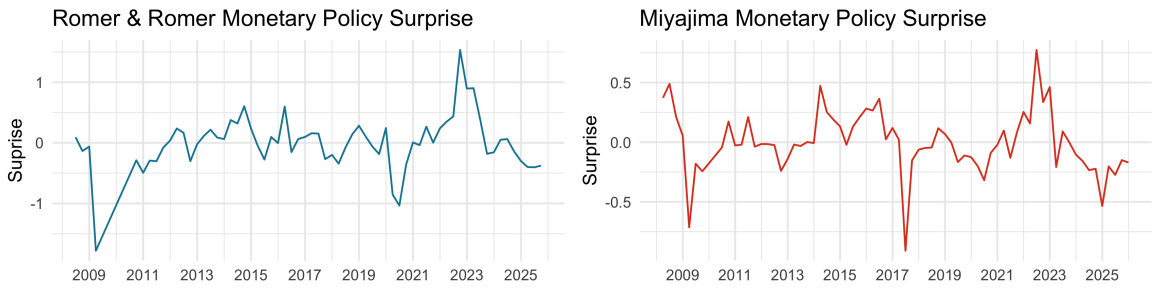

Figure A5: Theory-based monetary policy surprises: Romer–Romer and Miyajima measures, 2011–2025

Note: Two theory-based monetary policy surprise measures used as alternative identification strategies. The *Romer measure* (blue) is the residual from regressing the change in the SARB's repo rate forecast on its own internal projections for GDP growth and inflation, following the approach of Romer and Romer (2004) adapted to South Africa by Merrino (2022); a positive value indicates an unexpected tightening relative to the SARB's own information set. The *Miyajima measure* (red) is the residual from regressing the Bloomberg consensus forecast error for the policy rate on forecast errors for inflation and GDP growth (Miyajima, 2024); a positive value indicates the repo rate came in above what economic news alone would have predicted. Both series are aggregated to a quarterly frequency and purged of climate shock variation prior to use in the main regressions (see Equation 7).

A.2 Descriptive Statistics

Table A1: Summary statistics for key variables used in the analysis

	Variable	Mean	Median	SD	Min	Max	N
Climate shock							
	Population Precipitation Shock	-0.26	-0.48	1.88	-4.54	6.38	204
	Population Temperature Shock	0.96	0.88	0.69	-0.59	2.84	204
Lending rates							
	Corporate Mortgage Rate	8.43	9.07	2.04	4.06	11.35	212
	Corporate Secured Credit Rate	9.91	9.99	1.36	3.30	12.06	212
	Corporate Unsecured Credit Rate	8.71	8.62	1.74	5.95	12.59	212
	Household Mortgage Rate	9.30	9.75	1.41	6.56	11.67	212
	Household Secured Credit Rate	11.28	11.35	1.42	7.09	14.11	212
	Household Unsecured Credit Rate	14.52	14.54	1.96	10.22	18.37	212
Log of credit							
	Corporate Mortgages	25.12	25.27	0.54	23.95	25.94	212
	Corporate Secured Credit	24.17	24.22	0.37	23.38	24.95	212
	Corporate Unsecured Credit	25.75	25.73	0.28	25.04	26.42	212
	Household Mortgages	26.12	26.09	0.34	25.48	26.76	212
	Household Secured Credit	24.97	25.10	0.44	24.03	25.82	212
	Household Unsecured Credit	24.96	25.06	0.33	24.27	25.45	212
Monetary policy shock							
	Miyajima Surprise	0.00	-0.01	0.26	-0.91	0.77	208
	Romer Surprise	0.06	0.06	0.41	-1.04	1.53	204
	Target	0.03	0.01	0.13	-0.19	0.45	192

Note: Summary statistics computed over the full estimation sample (2011Q1–2025Q1) at the bank-quarter level. Credit volumes are expressed in log terms; lending rates are in percentage points; climate shocks are standardised deviations from the long-run calendar-month mean (see Equation 3); and monetary policy surprises are expressed in percentage points. The country risk factor, forward guidance factor, and central bank information factor are excluded from this

table to preserve space but are included in the main regressions. N denotes the number of non-missing bank-quarter observations.

B Credit growth results

B.1 Fixed effects

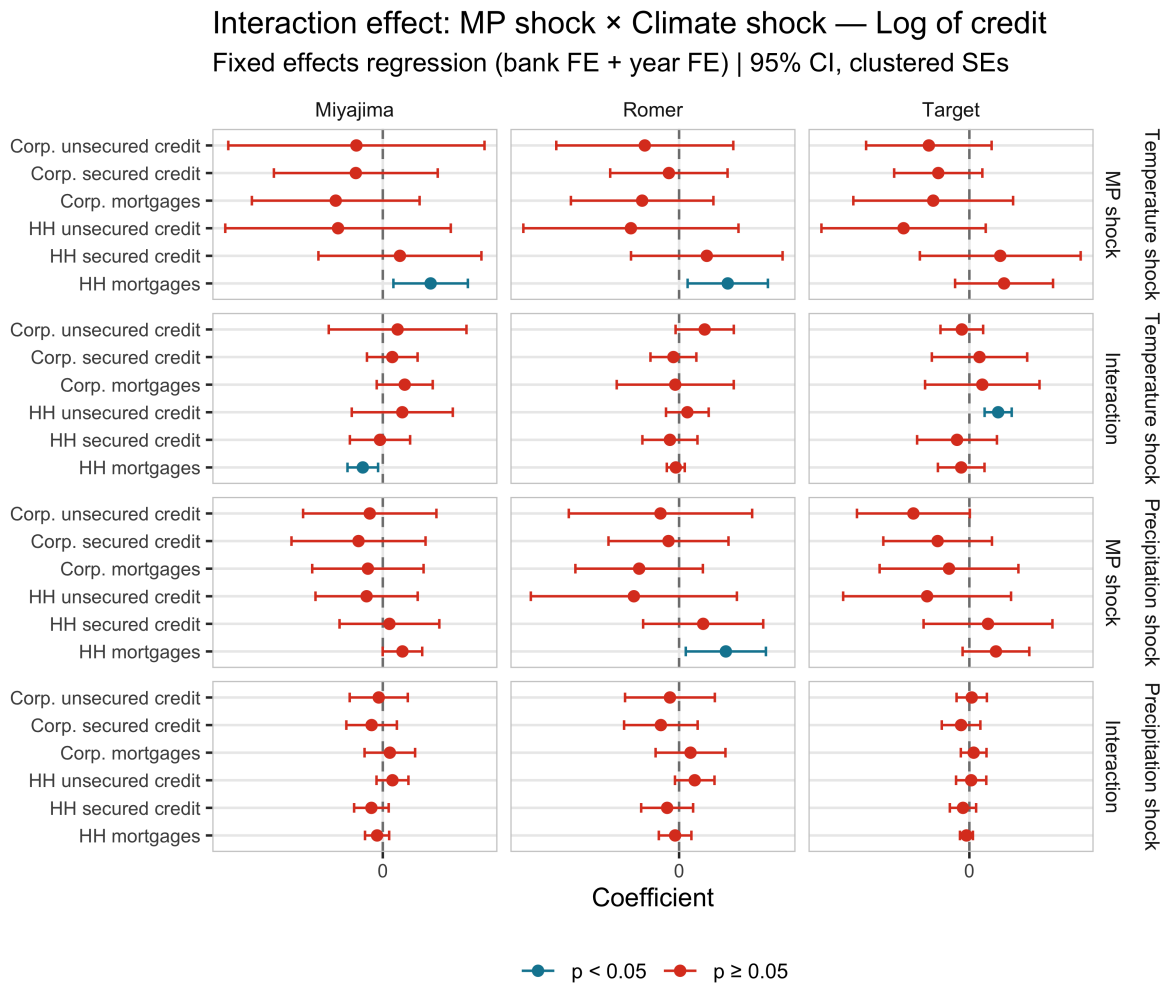

Figure A6: Interaction between climate and monetary policy shocks on credit growth: fixed effects estimates

Note: Each point shows the estimated coefficient $\hat{\beta}_3$ on the interaction term ClimateShock × MPShock from Equation 8, with 95% confidence intervals. Teal points indicate statistical significance at the 5% level; red points are insignificant. Rows correspond to individual log credit volume series disaggregated by borrower type and product; columns correspond

to the four monetary policy surprise measures retained after excluding the country risk and central bank information factors. All models include bank and quarter-of-year fixed effects with standard errors clustered at the bank level. Credit volumes are expressed in logarithms so coefficients represent proportional responses.

B.2 Local projections

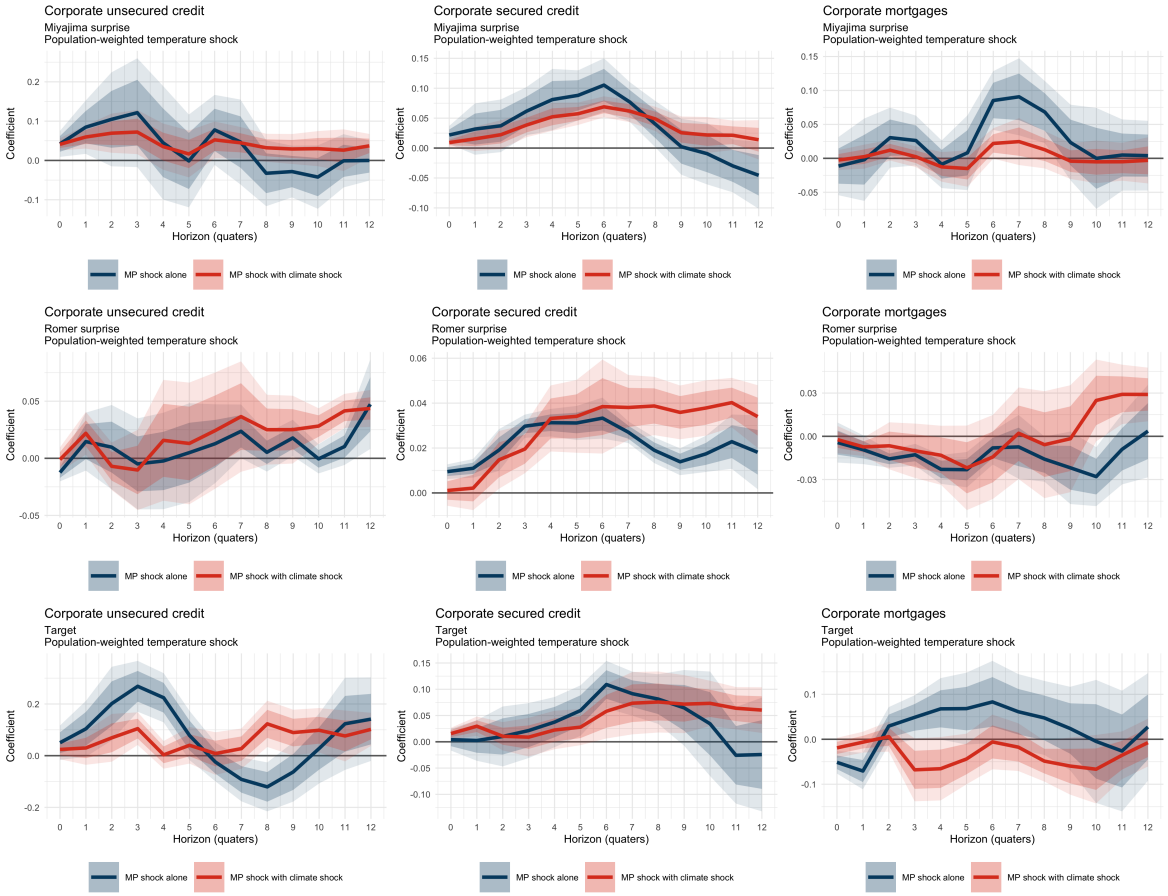

Figure A7: Impulse responses of corporate credit growth to monetary policy shocks: temperature shock interaction

Note: Impulse response functions (IRFs) from panel local projections estimated at horizons $h = 0, \ldots, 12$ quarters following Equation 9. The blue line shows the cumulative response of log corporate credit growth to the monetary policy shock alone ($\hat{\beta}_2$); the red line shows the response when the same shock coincides with the population-weighted temperature shock ($\hat{\beta}_2 + \hat{\beta}_3$). Shaded bands are 68% and 90% Driscoll-Kraay confidence intervals. Rows correspond to corporate unsecured, secured, and mortgage credit; columns correspond to the Miyajima, Romer, and Target monetary policy surprise measures.

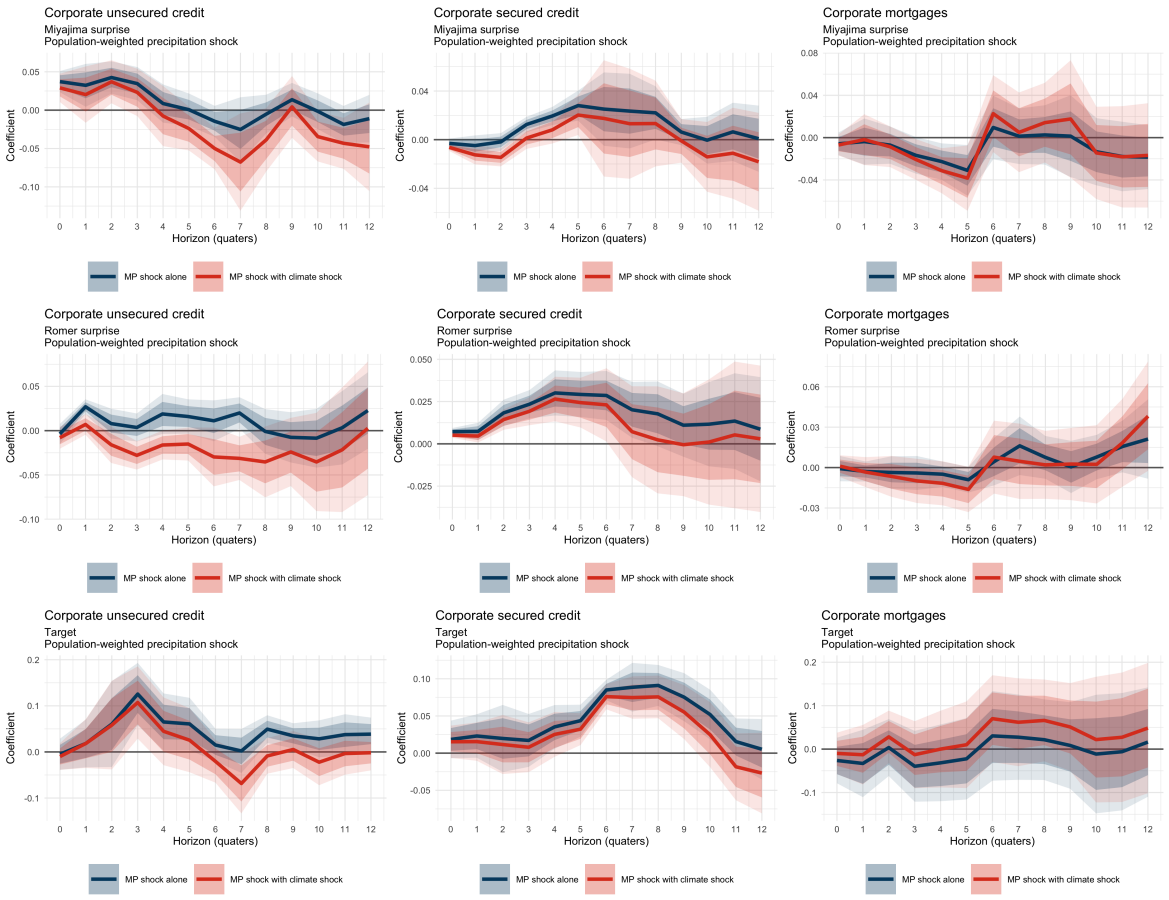

Figure A8: Impulse responses of corporate credit growth to monetary policy shocks: precipitation shock interaction

Note: As Figure A7 but for the population-weighted precipitation shock. The red line shows the corporate credit growth response when the monetary policy shock coincides with a precipitation shock ($\hat{\beta}_2 + \hat{\beta}_3$). Rows correspond to corporate unsecured, secured, and mortgage credit.

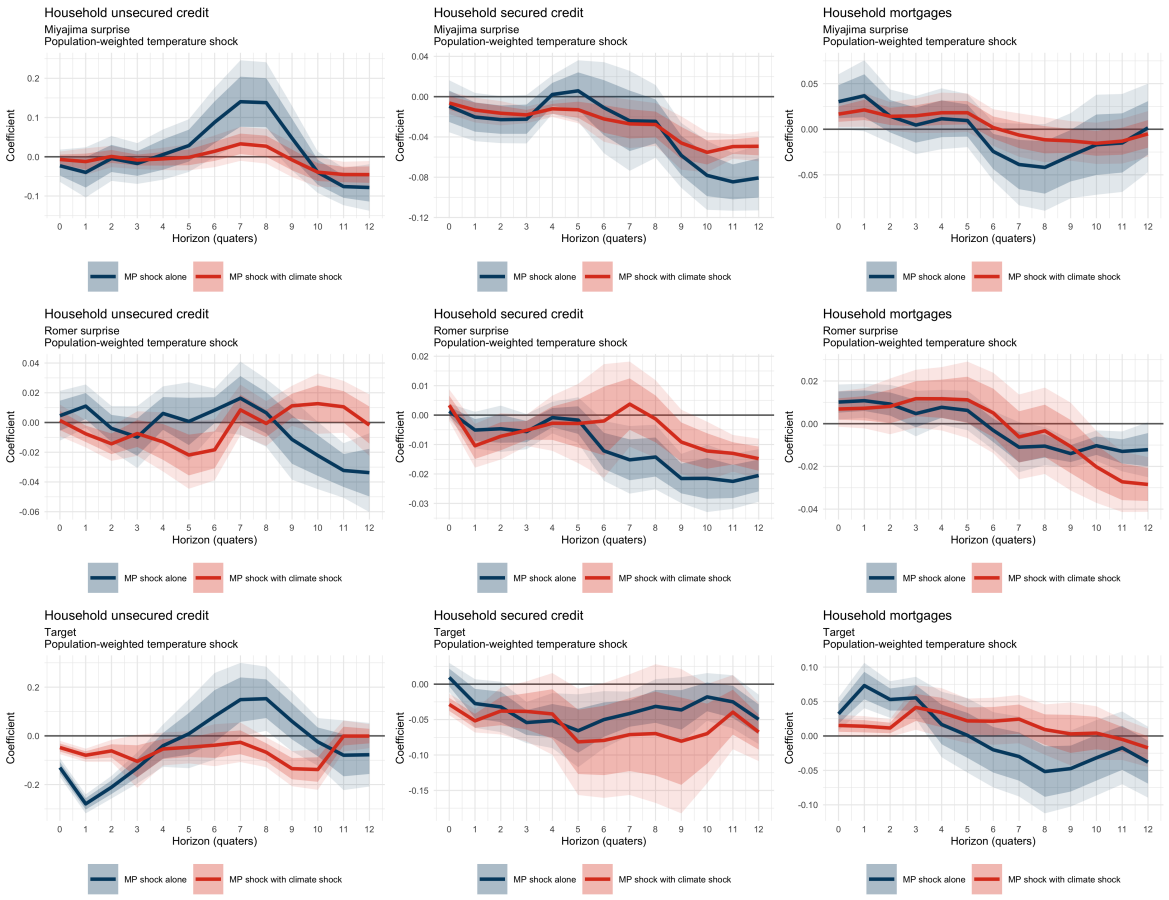

Figure A9: Impulse responses of household credit growth to monetary policy shocks: temperature shock interaction

Note: As Figure A7 but for household credit growth (unsecured, secured, and mortgage). The gap between the blue and red lines at each horizon represents the marginal effect of the temperature shock on the transmission of monetary policy to household credit volumes.

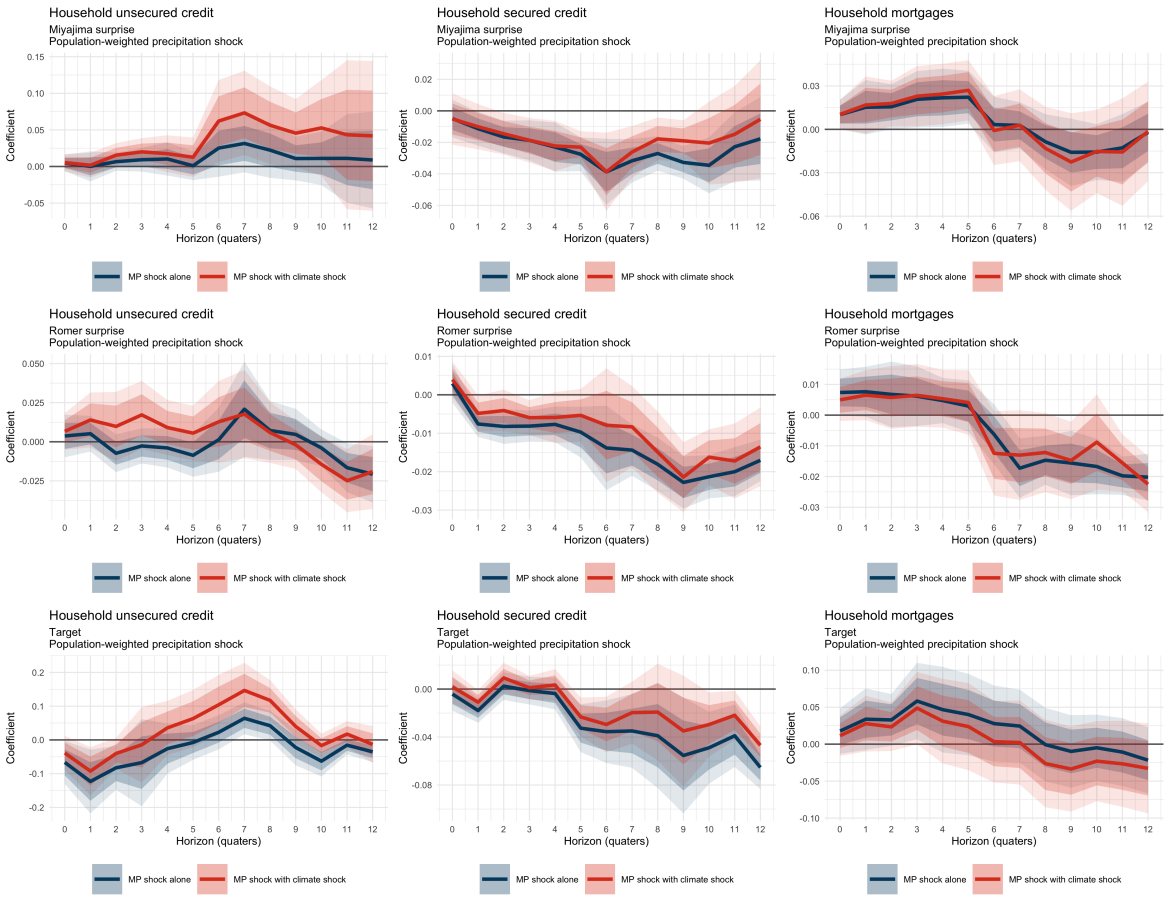

Figure A10: Impulse responses of household credit growth to monetary policy shocks: precipitation shock interaction

Note: As Figure A9 but for the population-weighted precipitation shock. Rows correspond to household unsecured, secured, and mortgage credit growth.

B.3 Asymmetric local projections

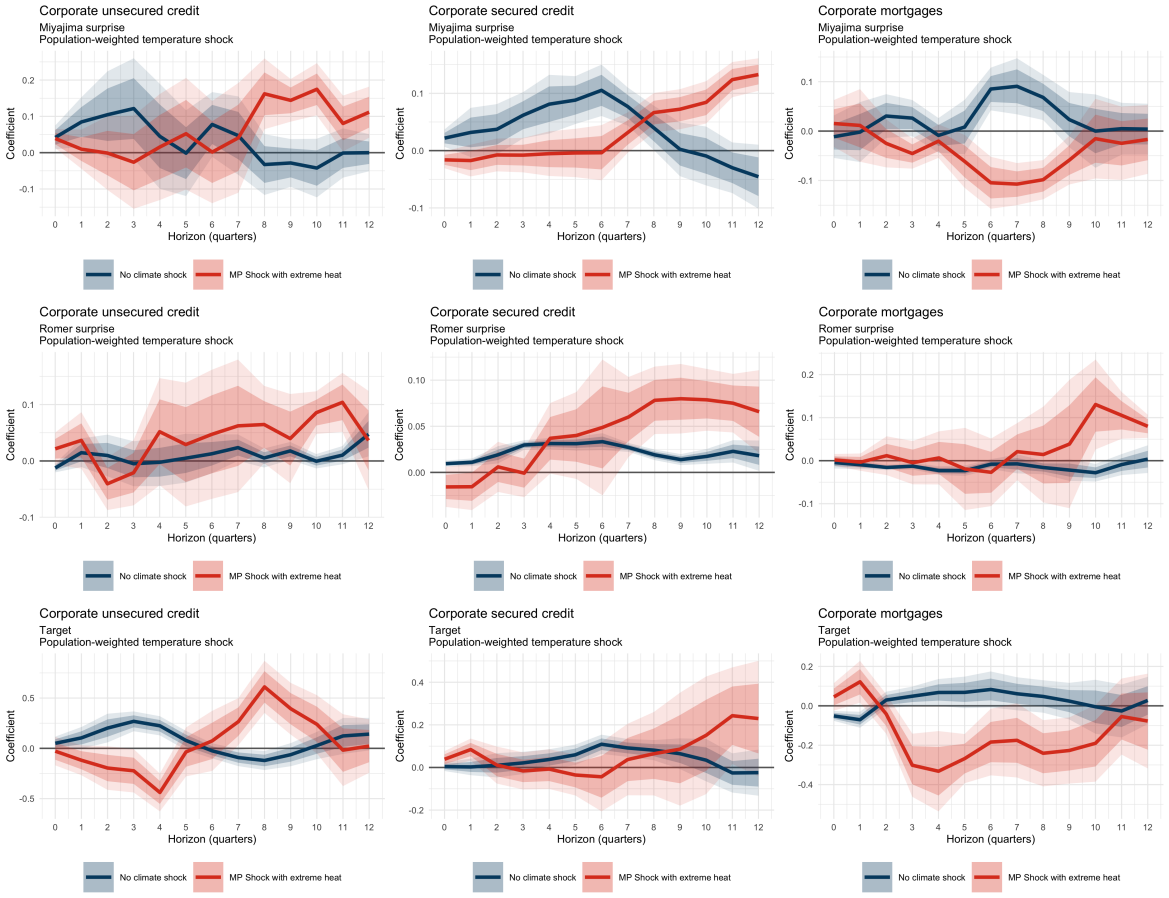

Figure A11: Asymmetric impulse responses of corporate credit growth under extreme temperature states

Note: Impulse response functions from asymmetric panel local projections. The blue line shows the cumulative response of log corporate credit growth to the monetary policy shock in the absence of a climate shock; the red line shows the response evaluated at an extreme positive temperature state (+3 SD). The gap between the two paths is the marginal climate amplification or dampening effect at that climate intensity. Shaded bands are 68% and 90% Driscoll-Kraay confidence intervals. Rows correspond to corporate unsecured, secured, and mortgage credit growth.

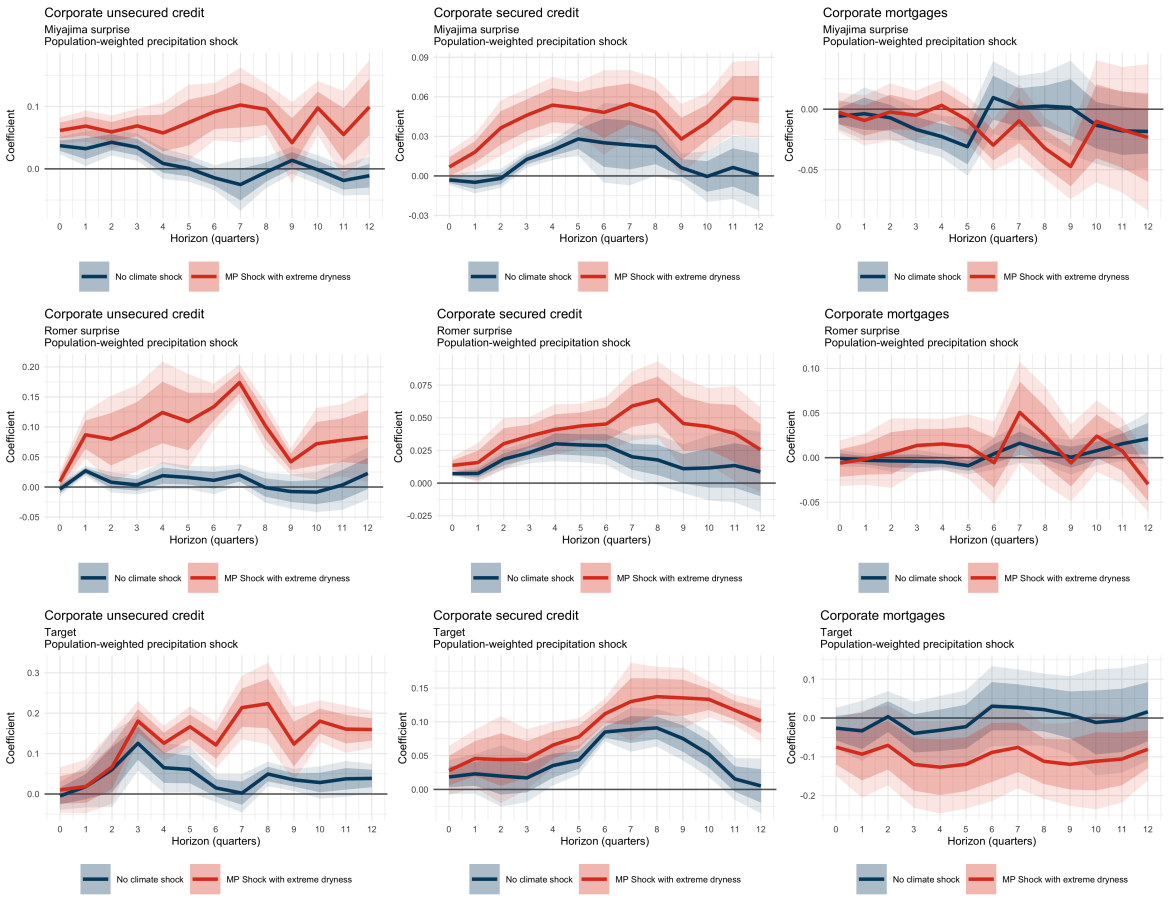

Figure A12: Asymmetric impulse responses of corporate credit growth under extreme precipitation states

Note: As Figure A11 but evaluated at an extreme negative precipitation state (−3 SD, i.e. drought conditions). A divergence between the red and blue lines indicates that drought conditions amplify or dampen the transmission of monetary policy to corporate credit volumes relative to the no-climate-shock baseline. Rows correspond to corporate unsecured, secured, and mortgage credit growth.

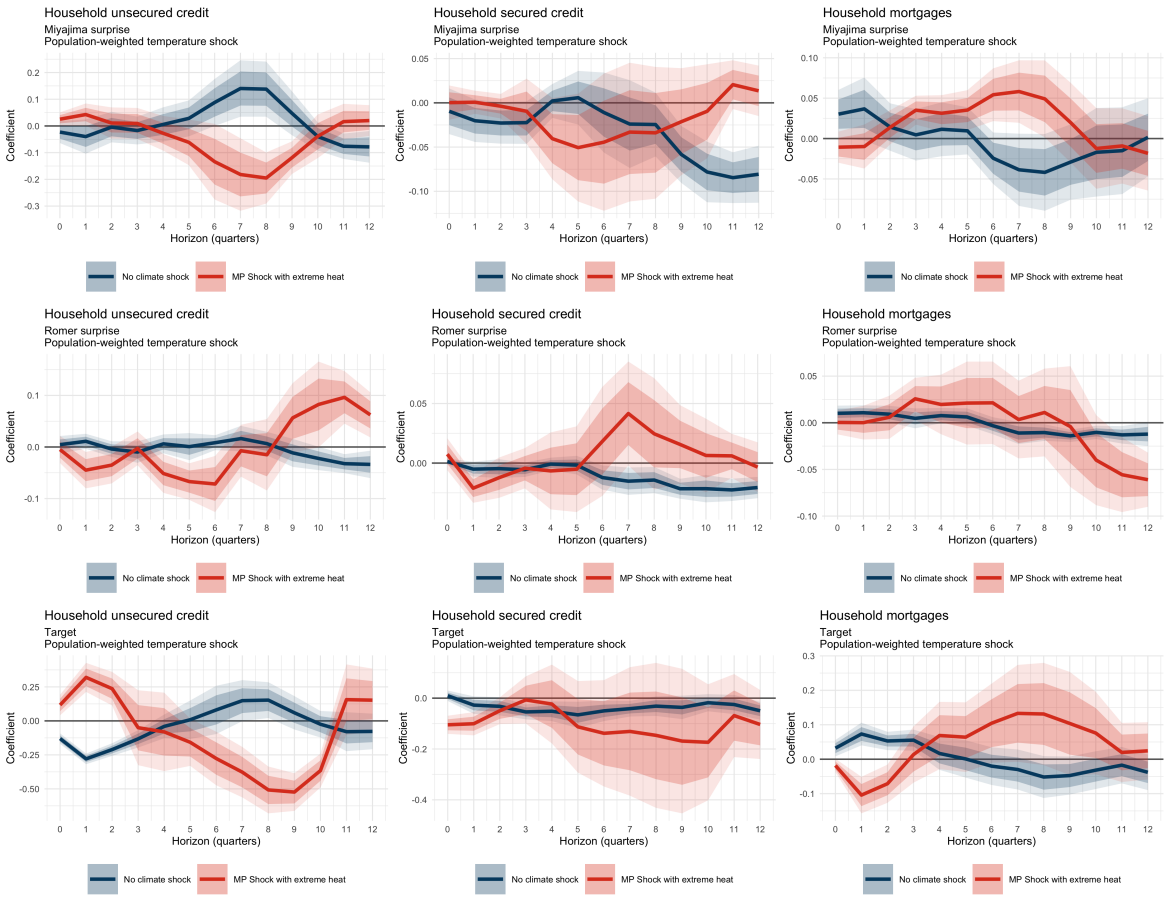

Figure A13: Asymmetric impulse responses of household credit growth under extreme temperature states

Note: As Figure A11 but for household credit growth (unsecured, secured, and mortgage). Compares the monetary policy transmission response in the absence of a climate shock against the response under an extreme positive temperature state (+3 SD).

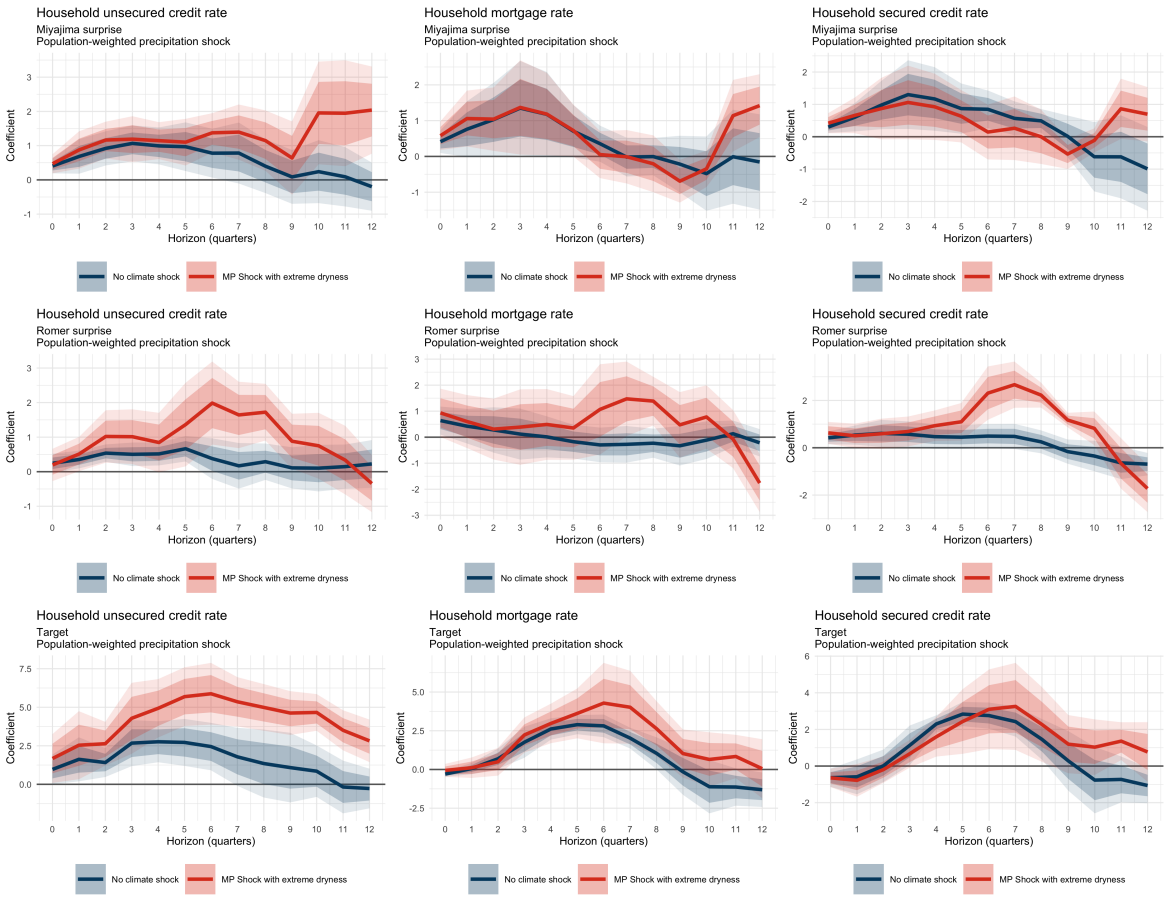

Figure A14: Asymmetric impulse responses of household credit growth under extreme precipitation states

Note: As Figure A13 but evaluated at an extreme negative precipitation state (−3 SD). Rows correspond to household unsecured, secured, and mortgage credit growth.